CAS SciFinder

Task History

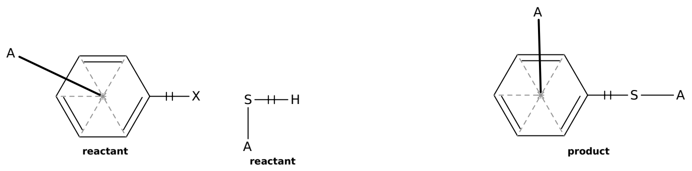
Initiating Search

October 13, 2025, 8:03 PM

Search:

Filtered By:

Yield:	**90-100%, 80-89%, 70-79%, <10%**
Reaction Mapping:	**Mapping Data Available**
Document Type:	**Journal**



Structure Match: **Substructure**

Search Tasks

Task	Result Type	View
Exported: Returned Reaction Results + Filters (8,617)	Reactions	View Results

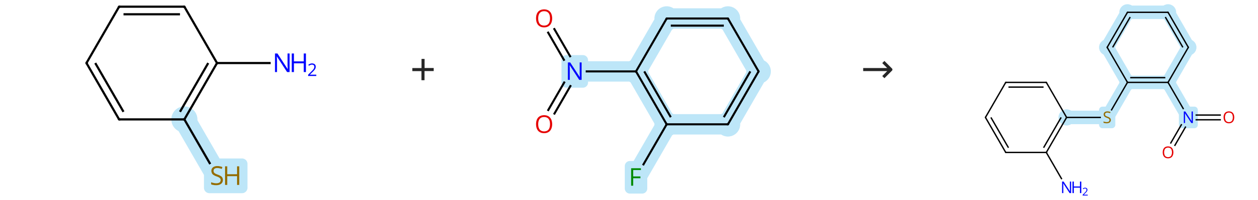
CAS SciFinder

Reactions (100)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (93) Suppliers (51)

31-614-CAS-43952833

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Acetonitrile; 15 min, rt

1.2 overnight, 110 °C

Experimental Protocols

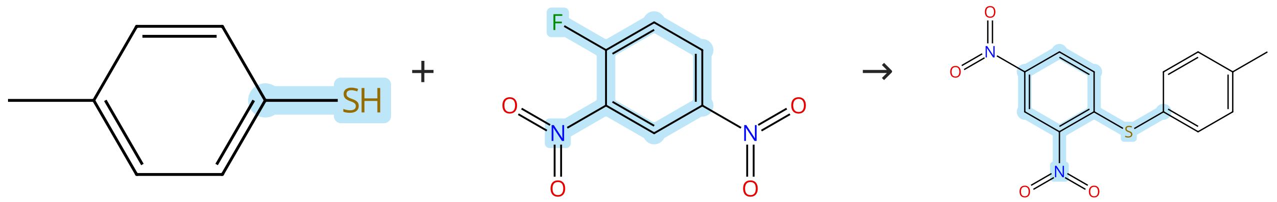
Synthesis and Reactivity of Six-Membered Cyclic Diaryl λ3-Bromanes and λ3-Chloranes

By: Bhattacherjee, Dhananjay; et al

Angewandte Chemie, International Edition (2025), 64(14), e202424559.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (56) Suppliers (75) Suppliers (3)

31-171-CAS-22372699

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Tripotassium phosphate, 18-Crown-6
Solvents: Tetrahydrofuran; 3 h, rt

Experimental Protocols

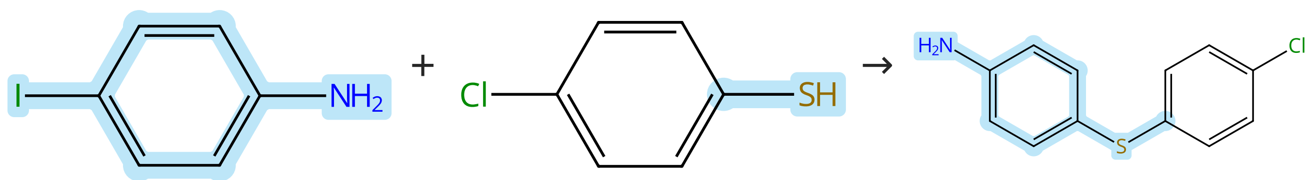
C-S coupling with nitro group as leaving group via simple inorganic salt catalysis

By: Xuan, Maojie; et al

Chinese Chemical Letters (2020), 31(1), 84-90.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (103) Suppliers (64) Suppliers (59)

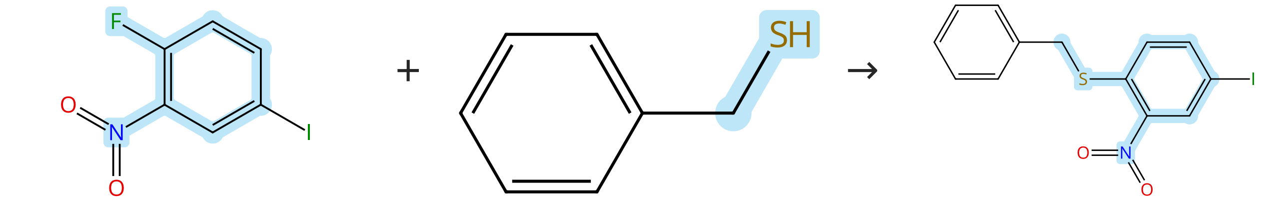
31-614-CAS-31632129	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cesium carbonate **Catalysts:** Copper oxide (CuO) **Solvents:** Toluene; 24 h, 110 °C
Experimental Protocols	

Rationale, synthesis and biological evaluation of substi tuted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (63)

Suppliers (53)

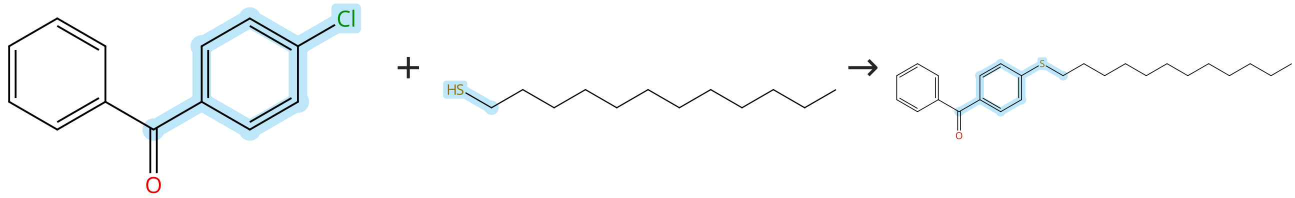
31-614-CAS-40567095	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 15 min, rt
1.2	-
Experimental Protocols	

Highly selective scalable electrosynthesis of 4-hydroxybenzo [e]-1,2,4-thiadiazine-1,1-dioxides

By: Winter, Johannes; et al

Cell Reports Physical Science (2024), 5(5), 101927.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (92)

Suppliers (57)

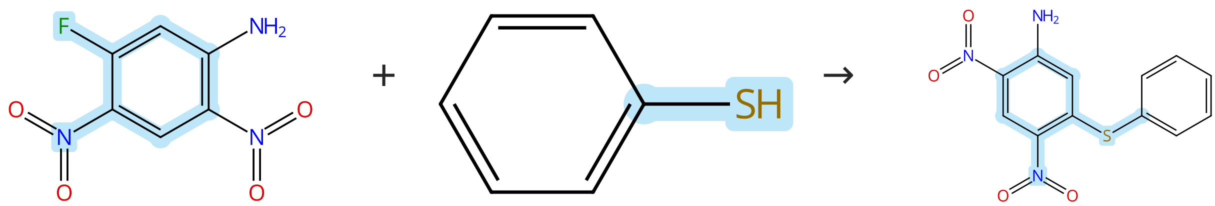
31-614-CAS-37149592	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 6 h, 0 °C
1.2	**Reagents:** Ammonium chloride **Solvents:** Water
Experimental Protocols	

Syntheses of aryl thioethers via aromatic substi tution of aryl halides at 0 to 25°C

By: Tsuzaki, Marina; et al

Chemical & Pharmaceutical Bulletin (2023), 71(7), 620-623.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (64)

Suppliers (44)

Suppliers (4)

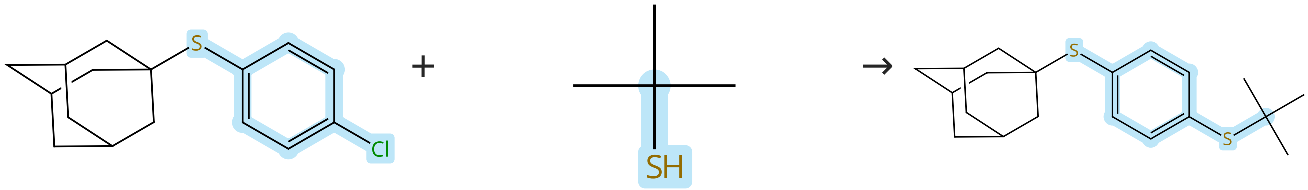
31-171-CAS-9084100	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Acetone; 16 h, rt
Experimental Protocols	

Design, synthesis and evaluation of novel 2,5,6-trisubstituted benzimidazoles targeting FtsZ as antitubercular agents

By: Park, Bora; et al

Bioorganic & Medicinal Chemistry (2014), 22(9), 2602-2612.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (50)

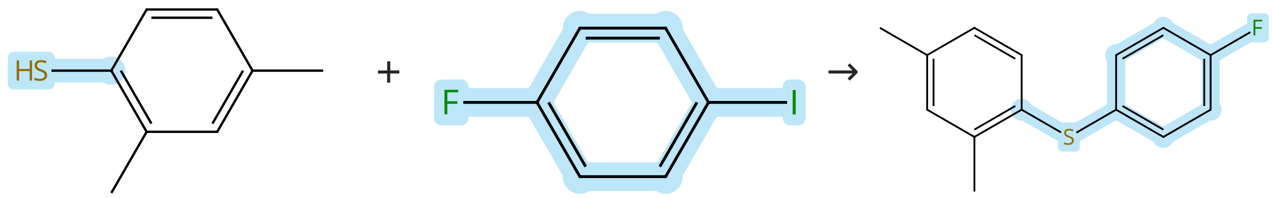
31-614-CAS-31329038	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium acetate **Catalysts:** (*SP*-5-34)-Chloro[1,1'-(9,9-dimethyl-9*H*-xanthene-4,5-diyl-κ*O*)bis[1,1-diphenylphosphine-κ*P*]](2-methylphenyl) nickel **Solvents:** Tetrahydrofuran; 2 h, rt
1.2	**Solvents:** Water; rt
Experimental Protocols	

Acetate Facilitated Nickel Catalyzed Coupling of Aryl Chlorides and Alkyl Thiols

By: Oechsner, Regina M.; et al

ACS Catalysis (2022), 12(4), 2233-2243.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (83) Supplier (1)

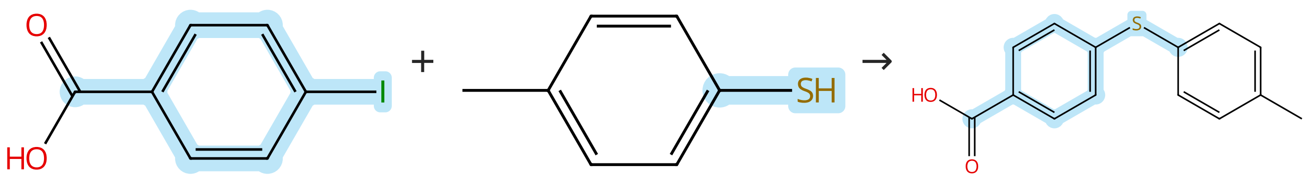
31-171-CAS-13863830	Steps: **1** Yield: **100%**
1.1	**Reagents:** 2,9-Dimethyl-1,10-phenanthroline, Sodium *tert*-butoxide **Solvents:** Toluene; 48 h, 110 °C
Experimental Protocols	

Copper- and palladium-containing perovskites: Catalysts for the Ullmann and Sonogashira reactions

By: Lohmann, Sophie; et al

Synlett (2005), (8), 1291-1295.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (95) Suppliers (56) Suppliers (9)

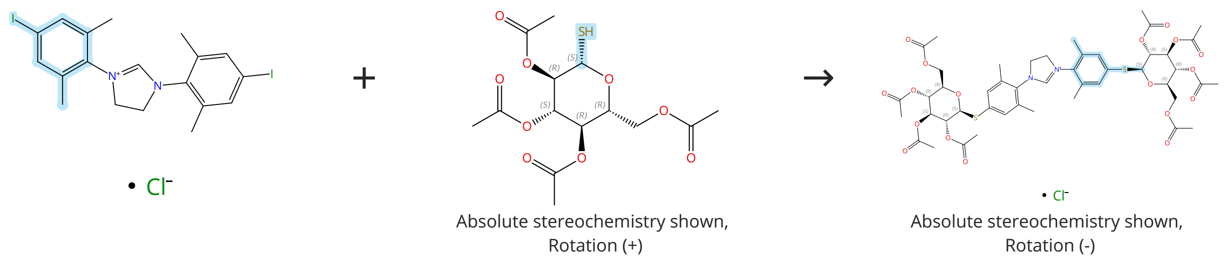
31-171-CAS-5581260	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diisopropylcarbodiimide **Catalysts:** 4-(Dimethylamino)pyridine **Solvents:** Dimethylformamide, Dichloromethane; 15 h, 50 °C
1.2	**Reagents:** Amberlite IRA 400 (borohydride bound), Borohydride (bound to IRA 400) **Catalysts:** Nickel, bis(2,2'-bipyridine-κ*N*[1],κ*N*[1'])dibromo- **Solvents:** Ethanol, Tetrahydrofuran; 15 h, 70 °C
1.3	**Reagents:** Trifluoroacetic acid **Solvents:** Dichloromethane; rt
Experimental Protocols	

Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (66)

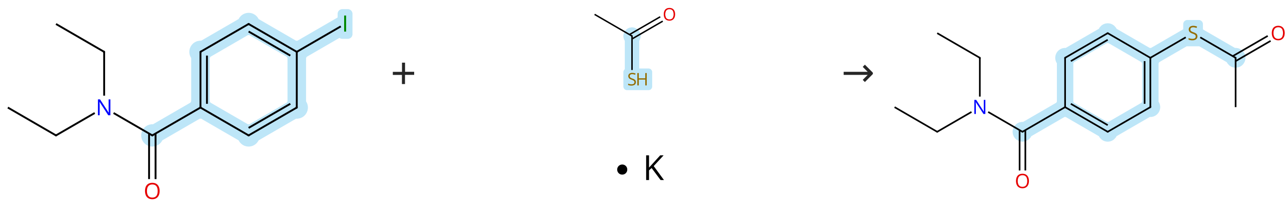
31-614-CAS-34373388	Steps: **1** Yield: **100%**
1.1	**Reagents:** Triethylamine **Catalysts:** [2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*][[5-(diphenylphosphino)-9,9-dimethyl-9*H*-xanthen-4-yl]diphenylphosphine-κ*P*](methanesulfonato-κ*O*)palladium **Solvents:** Tetrahydrofuran; 5 min, rt
1.2	**Solvents:** Tetrahydrofuran; 20 min, rt; 30 min, rt
Experimental Protocols	

Azoliums and Ag(I)-N-Heterocyclic Carbene Thioglycosides: Synthesis, Reactivity and Bioactivity

By: Ryzhakov, Dmytro; et al

European Journal of Organic Chemistry (2022), 2022(9), e202101499.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (45) Suppliers (84)

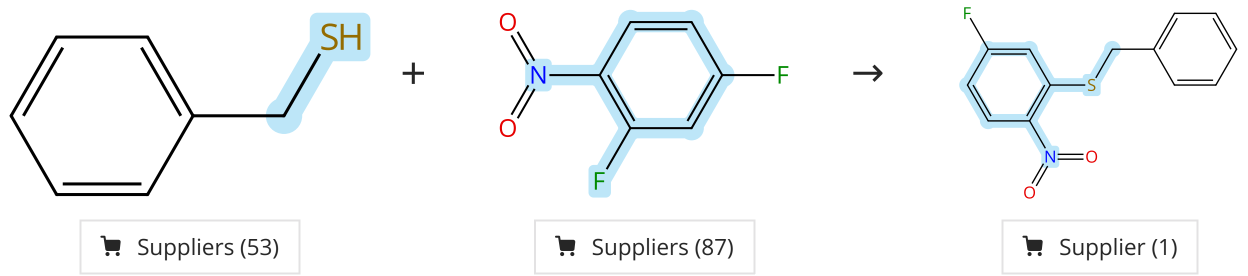
31-614-CAS-40340401	Steps: **1** Yield: **100%**
1.1	**Catalysts:** 1,10-Phenanthroline, Cuprous iodide **Solvents:** Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C
Experimental Protocols	

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-8818695 Steps: **1** Yield: **100%**

1.1 **Reagents:** Triethylamine
Solvents: Toluene; 0 - 5 °C; 15 min, 0 - 5 °C

1.2 0 - 5 °C

1.3 **Reagents:** Water

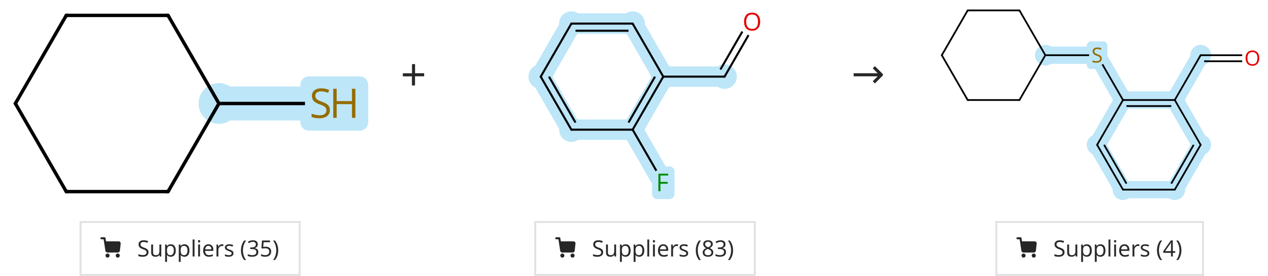
Experimental Protocols

Nonpolar Solvent a Key for Highly Regiose lective S_NAr Reaction in the Case of 2,4-Difluoronitrobenzene

By: Sythana, Suresh Kumar; et al

Organic Process Research & Development (2014), 18(7), 912-918.

Scheme 13 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-5089457 Steps: **1** Yield: **100%**

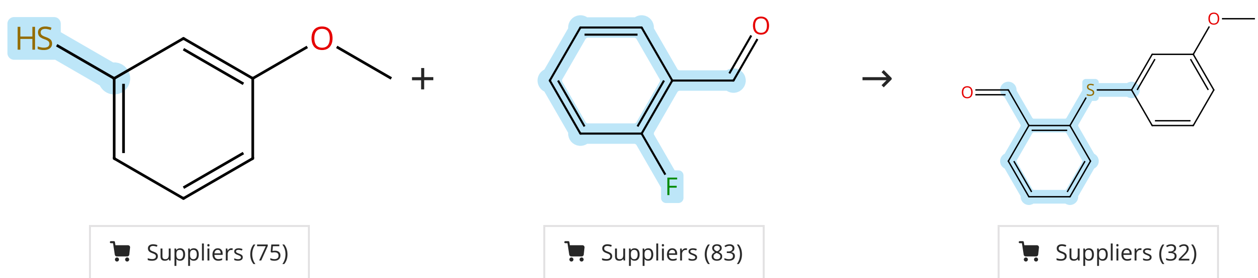
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 14 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-9149936 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

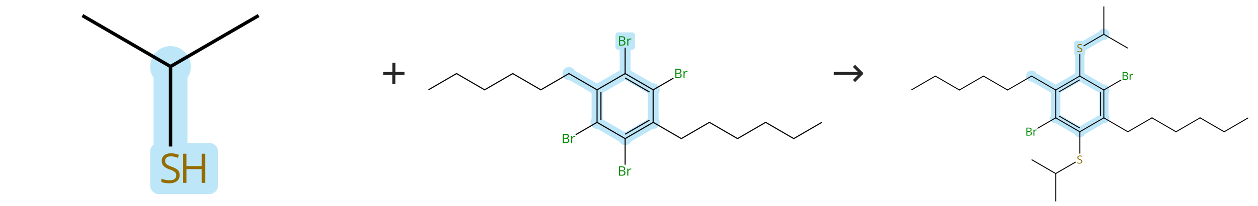
A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (55)

Suppliers (6)

31-171-CAS-5421626

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; rt; 30 min, rt

1.2 rt; overnight, 50 °C

Experimental Protocols

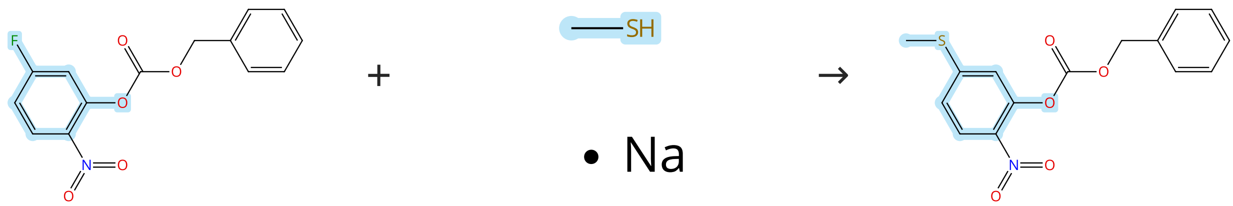
Highly Fluorinated Benzobisbenzothiophenes

By: Wang, Yongfeng; et al

Organic Letters (2008), 10(15), 3307-3310.

Scheme 16 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (5)

Suppliers (56)

31-171-CAS-9191755

Steps: **1** Yield: **100%**

1.1 **Reagents:** Pyridine
Solvents: Tetrahydrofuran, Water; 3 d, rt

Experimental Protocols

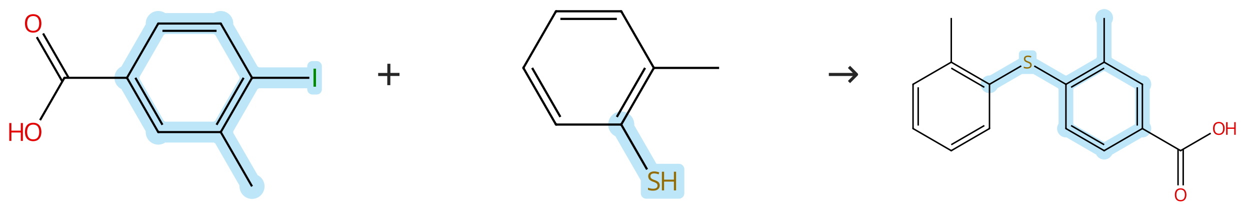
Identification, Optimization, and Pharmacology of Acylurea GHS-R1a Inverse Agonists

By: McCoull, William; et al

Journal of Medicinal Chemistry (2014), 57(14), 6128-6140.

Scheme 17 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (67)

Suppliers (58)

Suppliers (4)

31-171-CAS-9516278

Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dimethylformamide, Dichloromethane; 15 h, 50 °C

1.2 **Reagents:** Amberlite IRA 400 (borohydride bound), Borohydride (bound to IRA 400)
Catalysts: Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo-
Solvents: Ethanol, Tetrahydrofuran; 15 h, 70 °C

1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; rt

Experimental Protocols

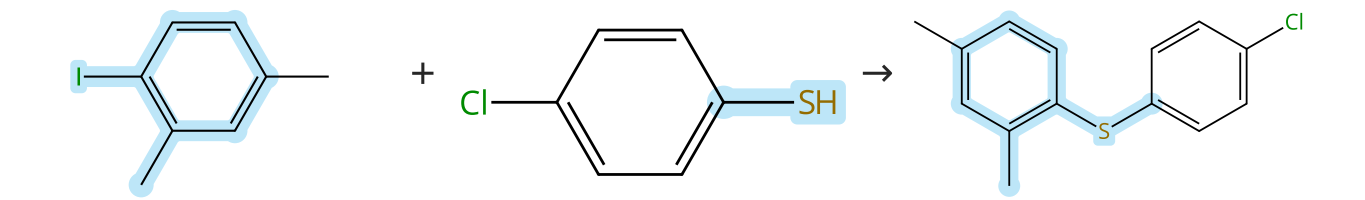
Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 18 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (71) | Suppliers (64) | Suppliers (4)

31-171-CAS-6433428

Steps: **1** Yield: **100%**

1.1 **Reagents:** Copper bromide (CuBr)
Catalysts: *N*'''-[*P*,*P*-Bis(dimethylamino)-*N*-ethylphosphinimyl]-*N*,*N*,*N*',*N*',*N*'',*N*''-hexamethylphosphorimidic triamide
Solvents: Toluene

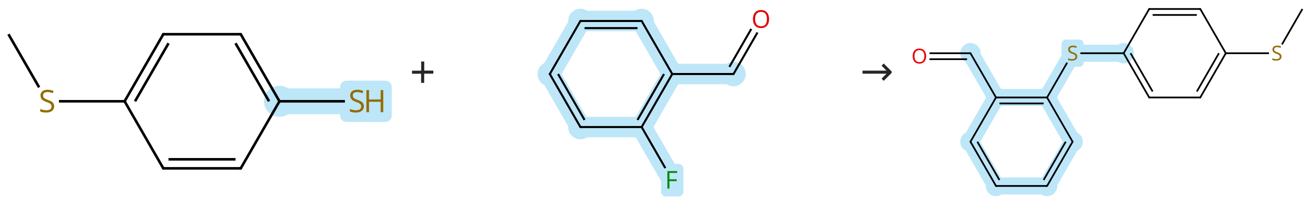
Phosphazene bases for the preparation of biaryl thioethers from aryl iodides and arenethiols

By: Palomo, Claudio; et al

Tetrahedron Letters (2000), 41(8), 1283-1286.

Scheme 19 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (64) | Suppliers (83) | Supplier (1)

31-171-CAS-15530713

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

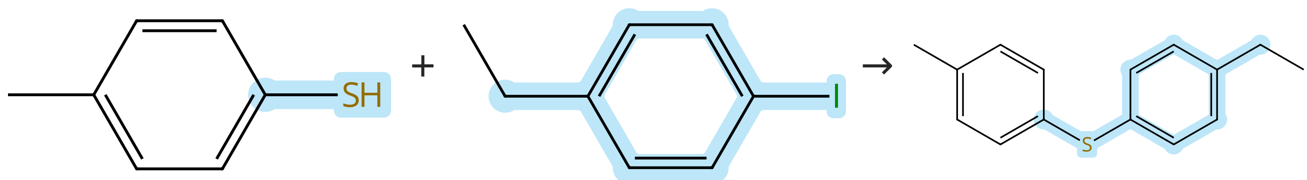
A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 20 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (56) | Suppliers (71) | Suppliers (5)

31-171-CAS-23873774

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C

1.2 **Catalysts:** Cuprous iodide; 12 h, 110 °C

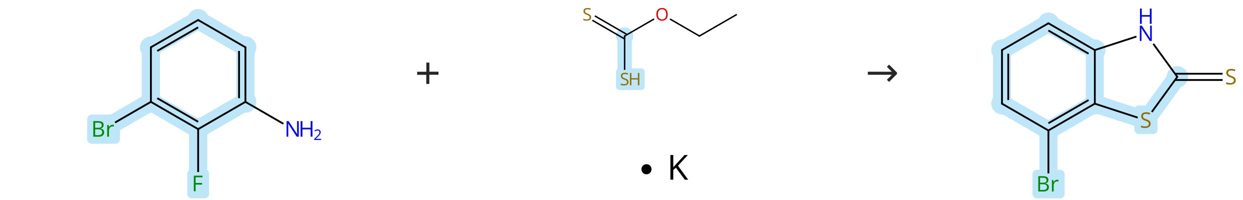
Experimental Protocols

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (70) Suppliers (57)

31-171-CAS-8470836 Steps: **1** Yield: **100%**

1.1 **Solvents:** Dimethylformamide; 3 h, 120 °C; 120 °C → rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 5, rt

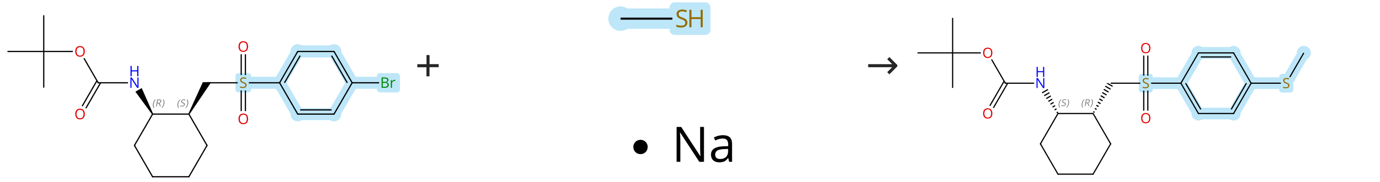
Experimental Protocols

Small molecules enhance functional O-mannosylation of Alpha-dystroglycan

By: Lv, Fengping; et al

Bioorganic & Medicinal Chemistry (2015), 23(24), 7661-7670.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



Relative stereochemistry shown

Suppliers (56)

Relative stereochemistry shown

31-171-CAS-13723082 Steps: **1** Yield: **100%**

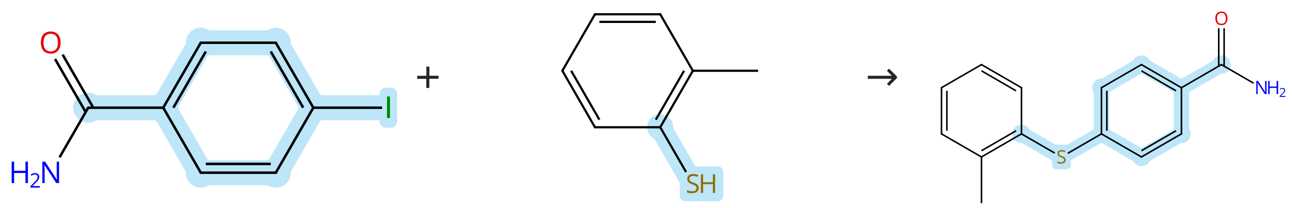
1.1 **Solvents:** Dimethylformamide

Novel sulfone-containing di- and trisubstituted cyclohexanes as potent CC chemokine receptor 2 (CCR2) antagonists

By: Cherney, Robert J.; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(13), 3418-3422.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (58) Supplier (1)

31-171-CAS-7607543 Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N*,*N*,*N'*,*N'*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide, Dichloromethane; 15 h, 45 °C

1.2 **Reagents:** Amberlite IRA 400 (borohydride bound), Borohydride (bound to IRA 400)
Catalysts: Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo-
Solvents: Ethanol, Tetrahydrofuran; 15 h, 70 °C

1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; rt

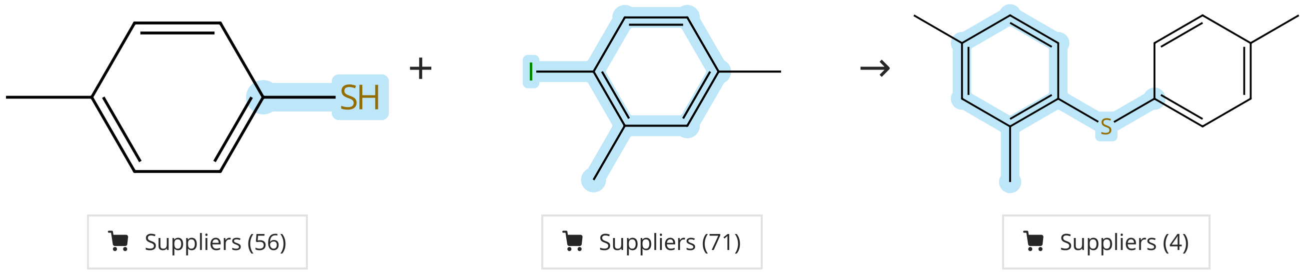
Experimental Protocols

Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 24 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-23874822 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C
1.2 **Catalysts:** Cuprous iodide; 12 h, 110 °C

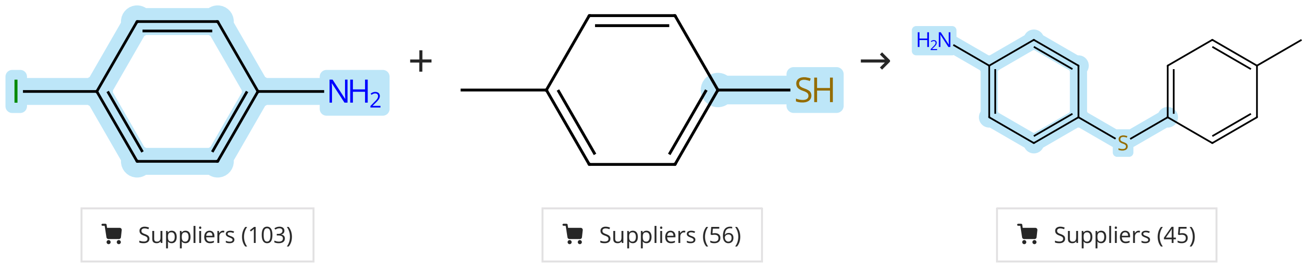
Experimental Protocols

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 25 (1 Reaction) Steps: **1** Yield: **100%**



31-614-CAS-31632127 Steps: **1** Yield: **100%**

1.1 **Reagents:** Cesium carbonate
Catalysts: Copper oxide (CuO)
Solvents: Toluene; 24 h, 110 °C

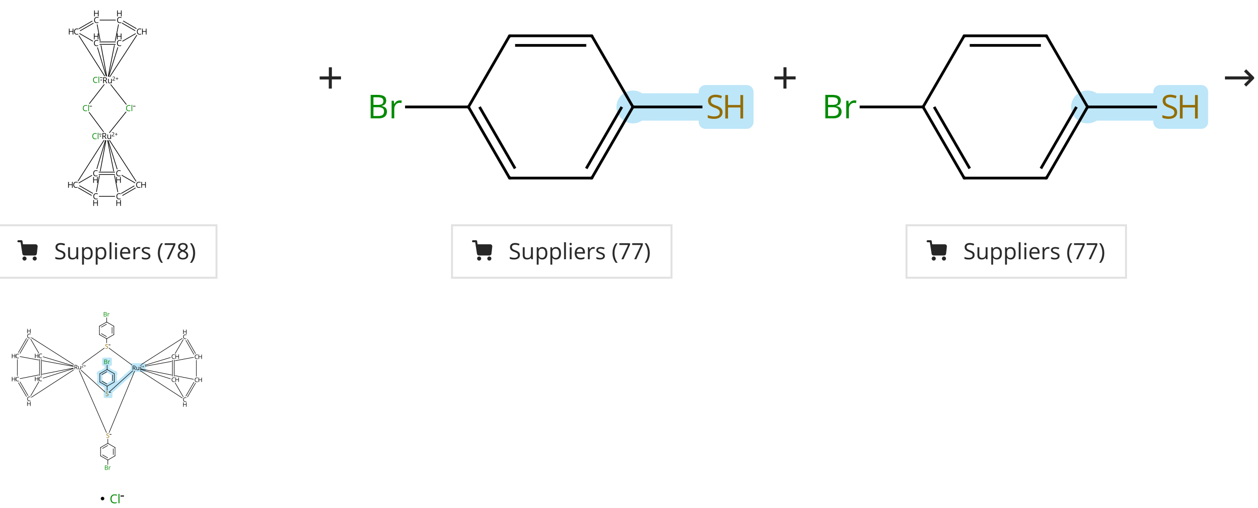
Experimental Protocols

Rationale, synthesis and biological evaluation of substi tuted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 26 (1 Reaction) Steps: **1** Yield: **100%**



31-562-CAS-12333297 Steps: **1** Yield: **100%**

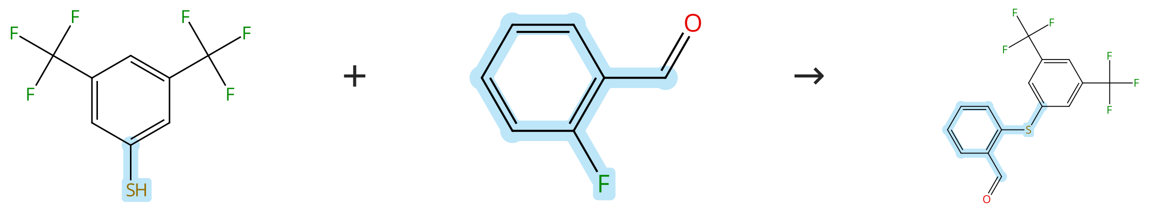
1.1 **Solvents:** Ethanol; reflux
1.2 3 h, reflux; reflux → 20 °C

Synthesis and structural characterisation of new cationic dinuclear ruthenium(II) thiolato complexes of the type [$Ru_2(\eta^6$-arene$)_2(\mu$-p-S-C_6H_4-Br$)_3]^+$

By: Cherioux, Frederic; et al

Inorganica Chimica Acta (2004), 357(3), 834-838.

Scheme 27 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (69) Suppliers (83) Suppliers (5)

31-171-CAS-4234308 Steps: **1** Yield: **100%**

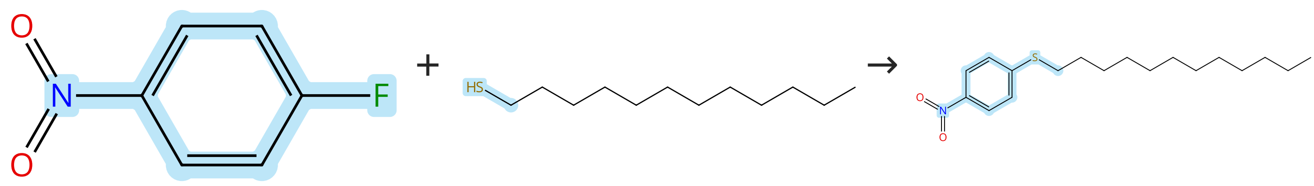
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 28 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (100) Suppliers (57) Suppliers (6)

31-171-CAS-984451 Steps: **1** Yield: **100%**

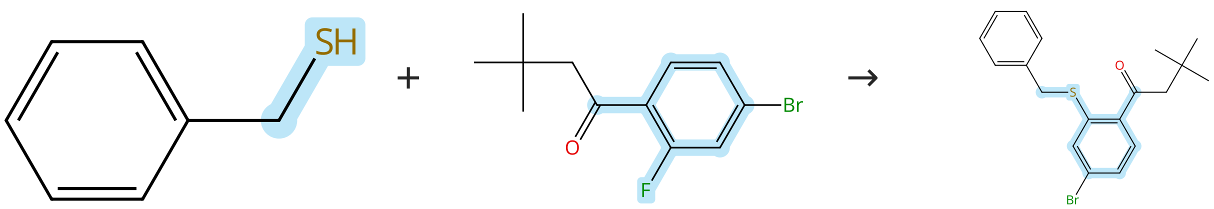
1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 45 min, 25 °C

Nucleophilic aromatic substitution reaction of nitroarenes with alkyl- or arylthio groups in dimethyl sulfoxide by means of cesium carbonate

By: Kondoh, Azusa; et al

Tetrahedron (2006), 62(10), 2357-2360.

Scheme 29 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (30) Supplier (1)

31-171-CAS-19716776	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; rt; 5 min, rt
1.2	**Solvents:** Tetrahydrofuran; 30 min, rt

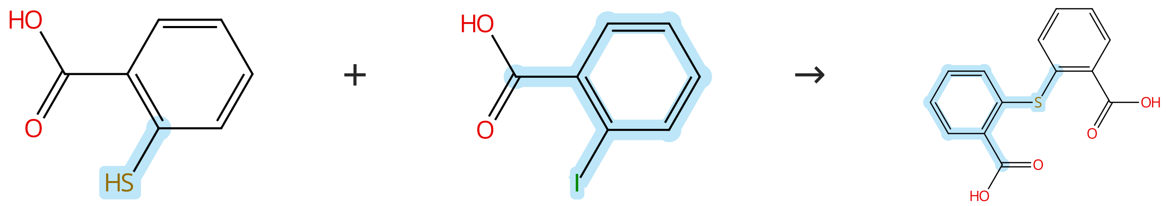
Experimental Protocols

Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 ($mGlu_4$)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 30 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (85) Suppliers (104) Suppliers (6)

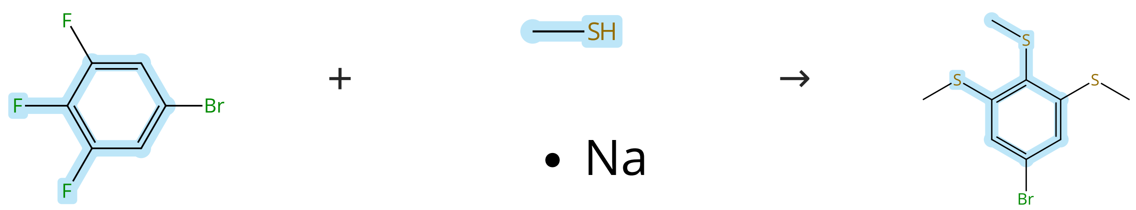
31-171-CAS-10255511	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Catalysts:** Copper **Solvents:** Water; reflux

Enzyme-promoted desymmetrization of bis(2-hydroxymethylphenyl) sulfoxide as a route to tridentate chiral catalysts

By: Rachwalski, Michal; et al

Tetrahedron: Asymmetry (2008), 19(17), 2096-2101.

Scheme 31 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (83) Suppliers (56) Supplier (1)

31-171-CAS-8791178	Steps: **1** Yield: **100%**
1.1	**Solvents:** 1,3-Dimethyl-2-imidazolidinone; rt; 3 h, rt; cooled

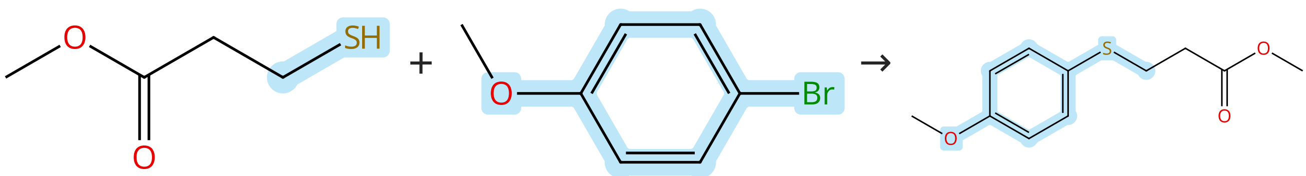
Experimental Protocols

Flexible Thioether-Ag(I) Interactions for Assembling Large Organic Ligands into Crystalline Networks

By: Huang, Guo; et al

Crystal Growth & Design (2009), 9(3), 1444-1451.

Scheme 32 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (75) Suppliers (71) Suppliers (6)

31-614-CAS-34820788	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Nickel bromide, Pyridine, hydriodide (1:1), 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine **Solvents:** Dimethylacetamide; 10 min, rt; 90 h, rt

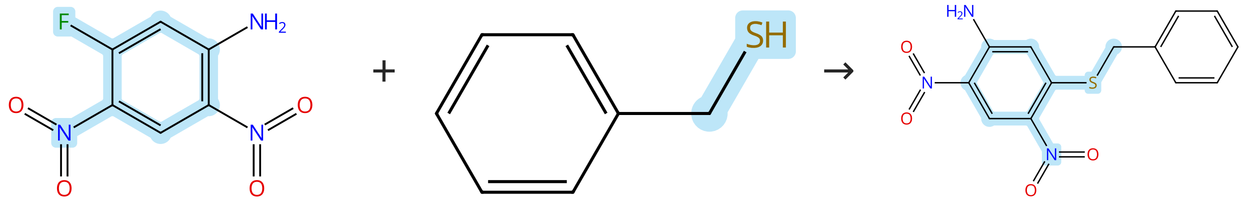
Experimental Protocols

Modulating the Surface and Photophysical Properties of Carbon Dots to Access Colloidal Photocatalysts for Cross-Couplings

By: Zhao, Zhouxiang; et al

ACS Catalysis (2022), 12(22), 13831-13837.

Scheme 33 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (64) Suppliers (53) Supplier (1)

31-171-CAS-13339631	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Acetone; 16 h, rt

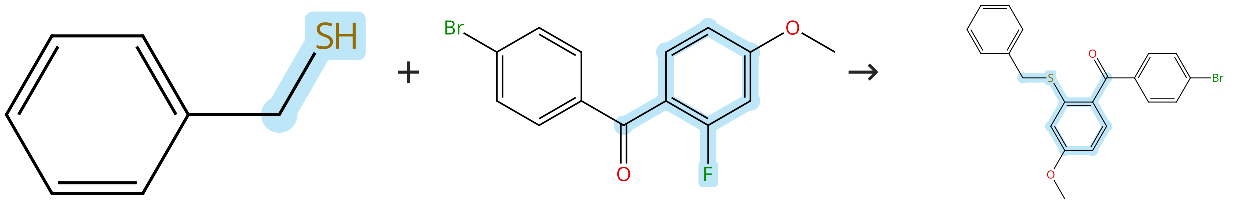
Experimental Protocols

Design, synthesis and evaluation of novel 2,5,6-trisubstituted benzimidazoles targeting FtsZ as antitubercular agents

By: Park, Bora; et al

Bioorganic & Medicinal Chemistry (2014), 22(9), 2602-2612.

Scheme 34 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (6)

31-171-CAS-14285711	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; rt; 30 min, rt
1.2	**Solvents:** Tetrahydrofuran; 1.5 h, rt
1.3	**Reagents:** Sodium bicarbonate, Ammonium chloride **Solvents:** Water; rt

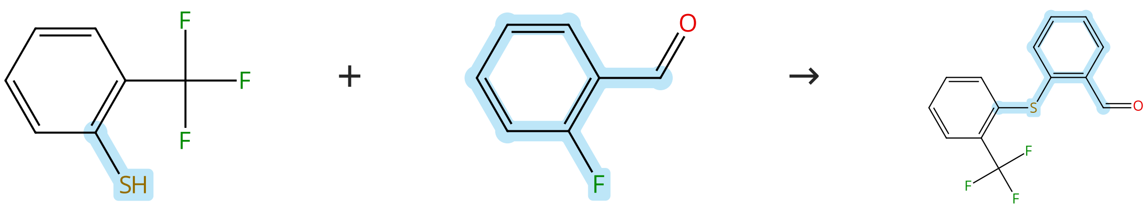
Experimental Protocols

Synthesis and Structure-Activity Studies of Novel Orally Active Non-Terpenoic 2,3-Oxidosqualene Cyclase Inhibitors

By: Dehmlow, Henrietta; et al

Journal of Medicinal Chemistry (2003), 46(15), 3354-3370.

Scheme 35 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (83) Supplier (1)

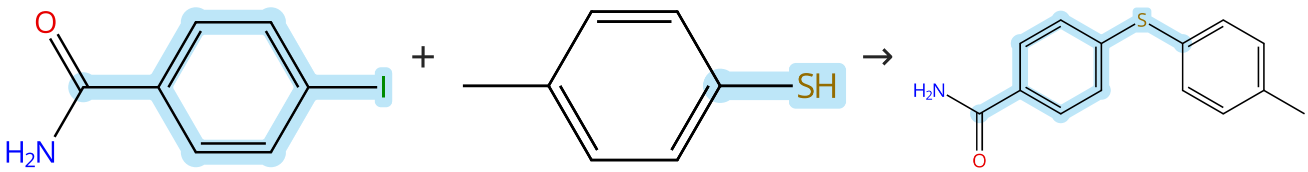
31-171-CAS-2141995	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 12 - 16 h, 50 - 80 °C

A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 36 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (56) Supplier (1)

31-171-CAS-5464827	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N*,*N*,*N'*,*N'*-tetramethyluronium hexafluorophosphate **Solvents:** Dimethylformamide, Dichloromethane; 15 h, 45 °C
1.2	**Reagents:** Amberlite IRA 400 (borohydride bound), Borohy dride (bound to IRA 400) **Catalysts:** Nickel, bis(2,2'-bipyridine-κN^{1},κ$N^{1'}$)dibromo- **Solvents:** Ethanol, Tetrahydrofuran; 15 h, 70 °C
1.3	**Reagents:** Trifluoroacetic acid **Solvents:** Dichloromethane; rt

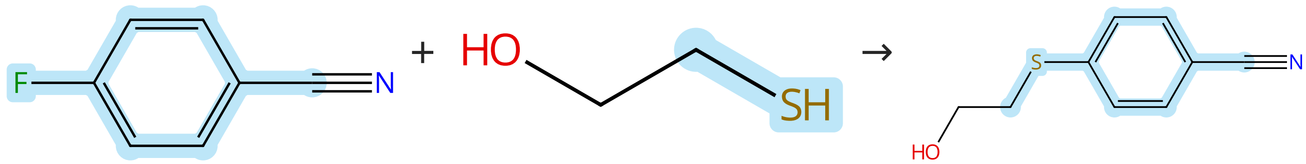
Experimental Protocols

Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 37 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (92) Suppliers (118) Suppliers (25)

31-171-CAS-266676	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Acetonitrile
1.2	**Solvents:** Ethyl acetate

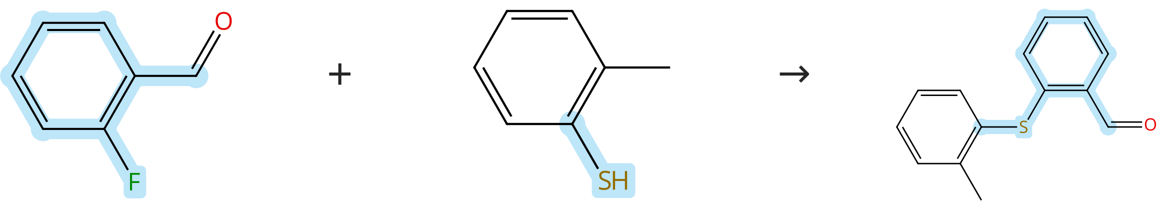
Experimental Protocols

Application of Base Cleavable Safety Catch Linkers to Solid Phase Library Production

By: Wade, Warren S.; et al

Journal of Combinatorial Chemistry (2000), 2(3), 266-275.

Scheme 38 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (83) Suppliers (58) Suppliers (5)

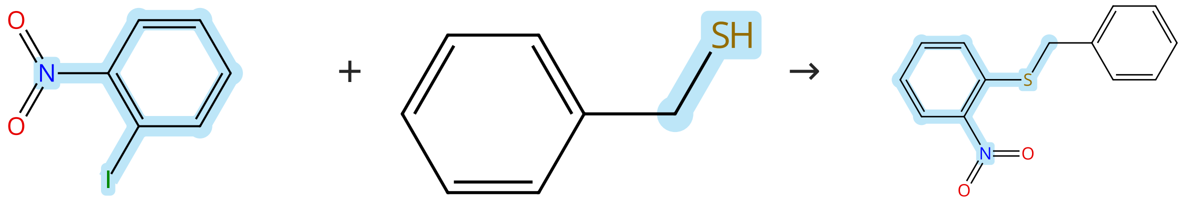
31-171-CAS-4605114	Steps: **1** Yield: **100%**
1.1 **Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 12 - 16 h, 50 - 80 °C	

A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 39 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (53) Suppliers (36)

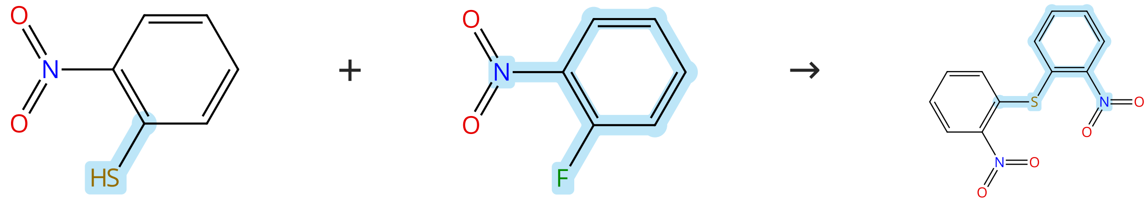
31-171-CAS-18817232	Steps: **1** Yield: **100%**
1.1 **Reagents:** Potassium carbonate **Catalysts:** Bis(L-prolinato-κ*N*,κ*O*)palladium **Solvents:** Ethanol; 6 h, 70 °C	

C-S Cross-Coupling Reaction Using a Recyclable Palladium Prolinate Catalyst under Mild and Green Conditions

By: Santos, Beatriz F.; et al

ChemistrySelect (2017), 2(28), 9063-9068.

Scheme 40 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (47) Suppliers (93) Suppliers (47)

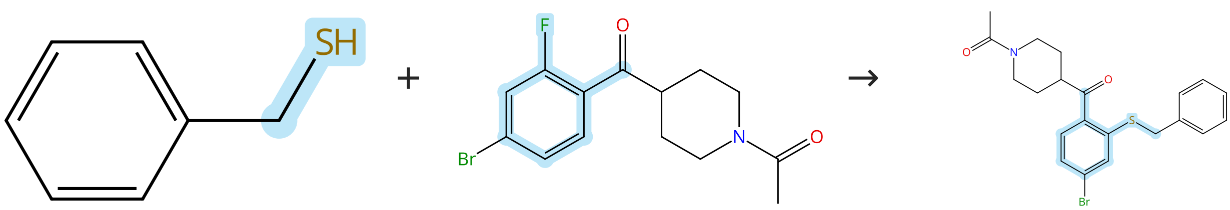
31-171-CAS-2674548	Steps: **1** Yield: **100%**
1.1 **Reagents:** Potassium carbonate **Solvents:** Dimethyl sulfoxide; 18 h, rt	
1.2 **Reagents:** Water; cooled	

Synthesis of Zirconium Complexes Containing the Tridentate Diamido Ligands $[(t\text{-Bu-}d_6\text{-N-}o\text{-}C_6H_4)_2S]^{2-}$ and $[(i\text{-PrN-}o\text{-}C_6H_4)_2S]^{2-}$

By: Graf, David D.; et al

Organometallics (1999), 18(5), 843-852.

Scheme 41 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (2) Supplier (1)

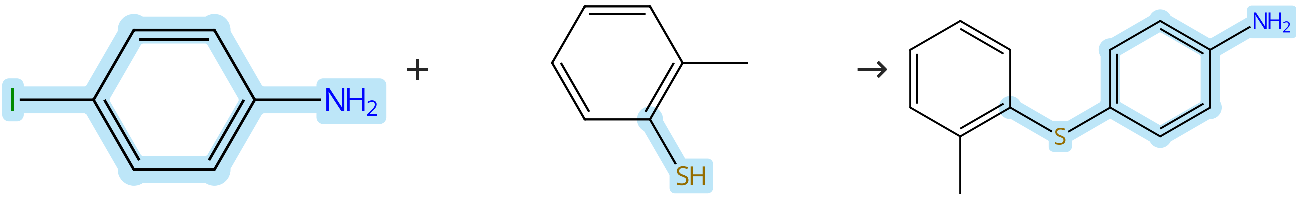
31-171-CAS-19716796	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; rt; 5 min, rt
1.2	**Solvents:** Tetrahydrofuran; 30 min, rt
Experimental Protocols	

Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 ($mGlu_4$)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 42 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (103) Suppliers (58) Suppliers (13)

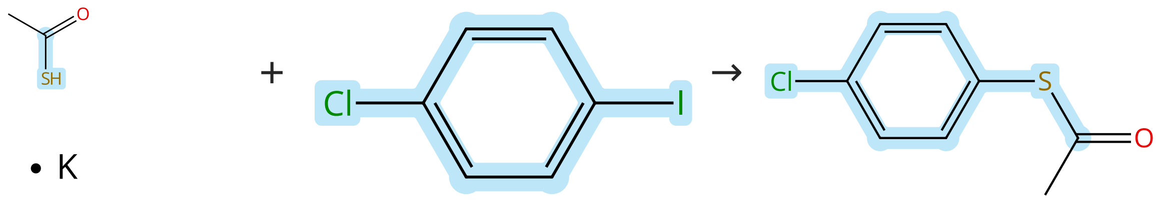
31-614-CAS-31632124	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cesium carbonate **Catalysts:** Copper oxide (CuO) **Solvents:** Toluene; 24 h, 110 °C
Experimental Protocols	

Rationale, synthesis and biological evaluation of substi tuted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 43 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (84) Suppliers (84) Suppliers (12)

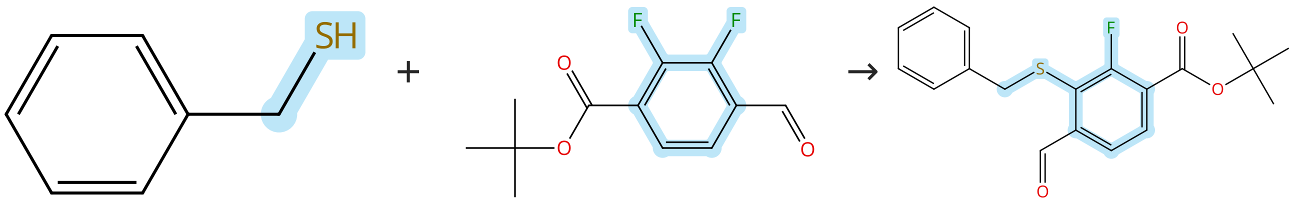
31-614-CAS-40340397	Steps: **1** Yield: **100%**
1.1	**Catalysts:** 1,10-Phenanthroline, Cuprous iodide **Solvents:** Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C
Experimental Protocols	

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 44 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (6) Suppliers (3)

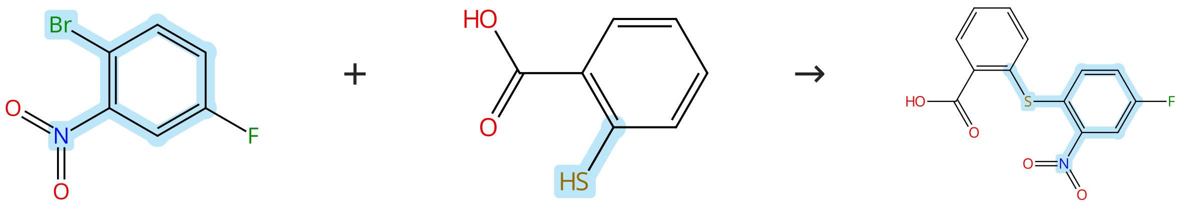
31-171-CAS-12256345	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; 5 min, rt; rt → - 30 °C
1.2	15 min, -30 °C; 30 min, - 30 °C
1.3	**Reagents:** Water
Experimental Protocols	

Discovery of Novel Allosteric Mitogen-Activated Protein Kinase Kinase (MEK) 1,2 Inhibitors Possessing Bidentate Ser212 Interactions

By: Heald, Robert A.; et al

Journal of Medicinal Chemistry (2012), 55(10), 4594-4604.

Scheme 45 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (92) Suppliers (85) Suppliers (4)

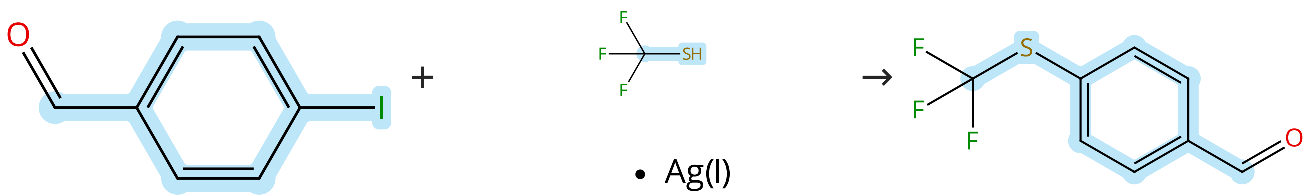
31-171-CAS-20665816	Steps: **1** Yield: **100%**
1.1	**Reagents:** Triethylamine, *N*-Methyl-2-pyrrolidone **Catalysts:** 1,1-Bis(diphenylphosphino)ferrocene, Tris (dibenzylideneacetone)dipalladium; 28 h, 85 °C

Preparation of the Serotonin Transporter PET Radiotracer 2-({2-[(Dimethylamino)methyl]phenyl}thio)-5-[^{18}F]fluoroaniline (4-[^{18}F]ADAM): Probing Synthetic and Radiosy nthetic Methods

By: Milicevic Sephton, Selena; et al

Synthesis (2019), 51(23), 4374-4384.

Scheme 46 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (93) Suppliers (51) Suppliers (69)

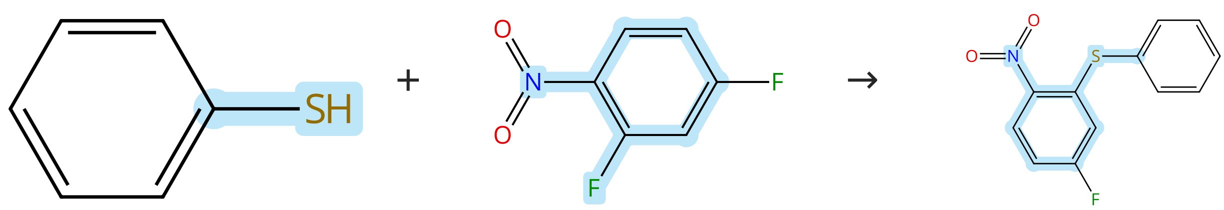
31-614-CAS-31383911	Steps: **1** Yield: **100%**
1.1	**Catalysts:** [2-[Bis(tricyclo[3.3.1.1^{3,7}]dec-1-yl)phosphino-κ*P*]-*N*, *N*-dimethylbenzenamine]chlorogold **Solvents:** 1,2-Dichloroethane; 1 min, rt
1.2	**Reagents:** Silver hexafluoroantimonate; 1 min, rt; 1.5 h, rt → 70 °C
Experimental Protocols	

Gold (I/III)-Catalyzed Trifluoromethylthiolation and Trifluoro methylselenolation of Organohalides

By: Mudshinge, Sagar R.; et al

Angewandte Chemie, International Edition (2022), 61(12), e202115687.

Scheme 47 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (44) Suppliers (87) Suppliers (2)

31-171-CAS-15206137	Steps: **1** Yield: **100%**
1.1	**Reagents:** Triethylamine **Solvents:** Toluene; 0 - 5 °C; 15 min, 0 - 5 °C
1.2	0 - 5 °C
1.3	**Reagents:** Water

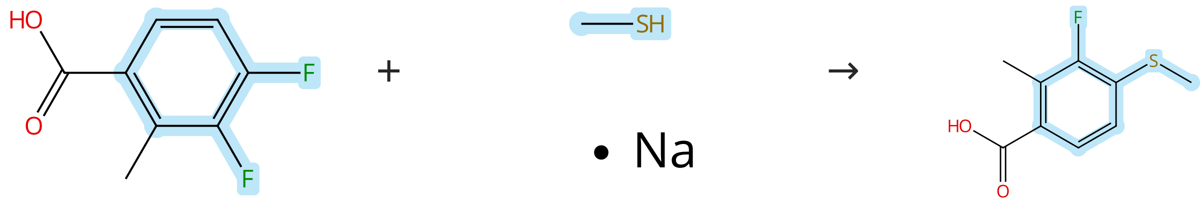
Experimental Protocols

Nonpolar Solvent a Key for Highly Regiose lective S_NAr Reaction in the Case of 2,4-Difluoronitrobenzene

By: Sythana, Suresh Kumar; et al

Organic Process Research & Development (2014), 18(7), 912-918.

Scheme 48 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (64) Suppliers (56) Suppliers (18)

31-171-CAS-8695146	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium hydroxide **Solvents:** Dimethyl sulfoxide; 30 min, rt
1.2	4 h, 55 - 60 °C; 60 °C → 0 °C
1.3	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 1 - 2

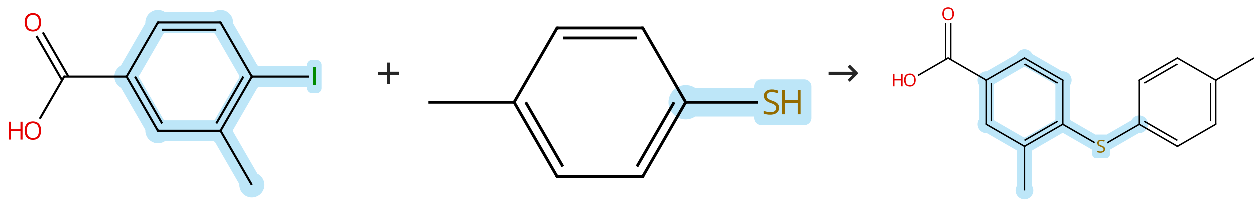
Experimental Protocols

Discovery of a Novel Class of Highly Potent, Selective, A TP-Competitive, and Orally Bioavailable Inhibitors of the Mammalian Target of Rapamycin (mTOR)

By: Takeuchi, Craig S.; et al

Journal of Medicinal Chemistry (2013), 56(6), 2218-2234.

Scheme 49 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (56) Suppliers (4)

31-171-CAS-7355394	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diisopropylcarbodiimide **Catalysts:** 4-(Dimethylamino)pyridine **Solvents:** Dimethylformamide, Dichloromethane; 15 h, 50 °C
1.2	**Reagents:** Amberlite IRA 400 (borohydride bound), Borohy dride (bound to IRA 400) **Catalysts:** Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo- **Solvents:** Ethanol, Tetrahydrofuran; 15 h, 70 °C
1.3	**Reagents:** Trifluoroacetic acid **Solvents:** Dichloromethane; rt

Experimental Protocols

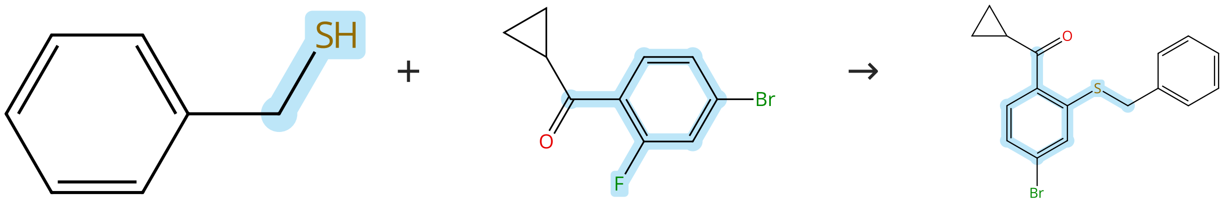
Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (53)

Suppliers (52)

31-171-CAS-19716778

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; rt; 5 min, rt

1.2 **Solvents:** Tetrahydrofuran; 30 min, rt

Experimental Protocols

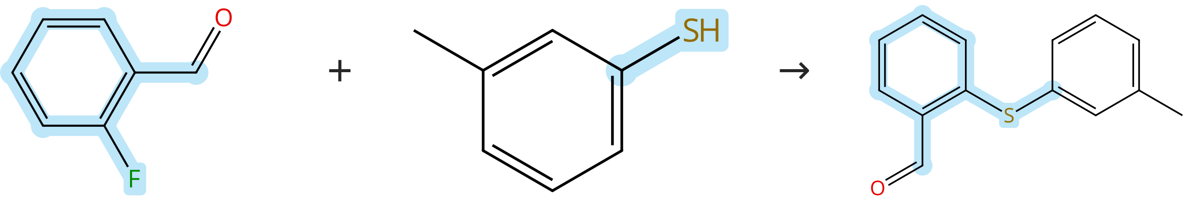
Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu$_4$)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 51 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (83)

Suppliers (56)

Suppliers (4)

31-171-CAS-289509

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

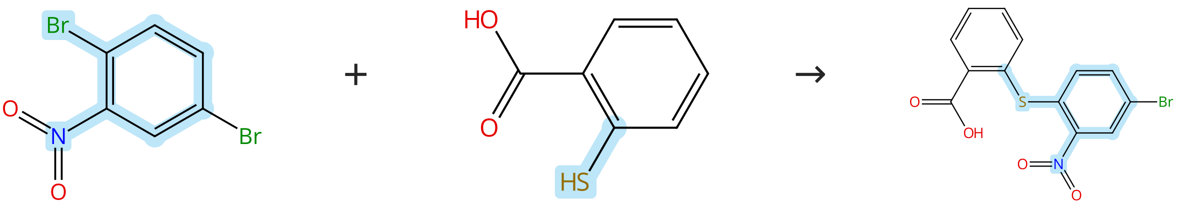
A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 52 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (91)

Suppliers (85)

Suppliers (3)

31-171-CAS-20665818

Steps: **1** Yield: **100%**

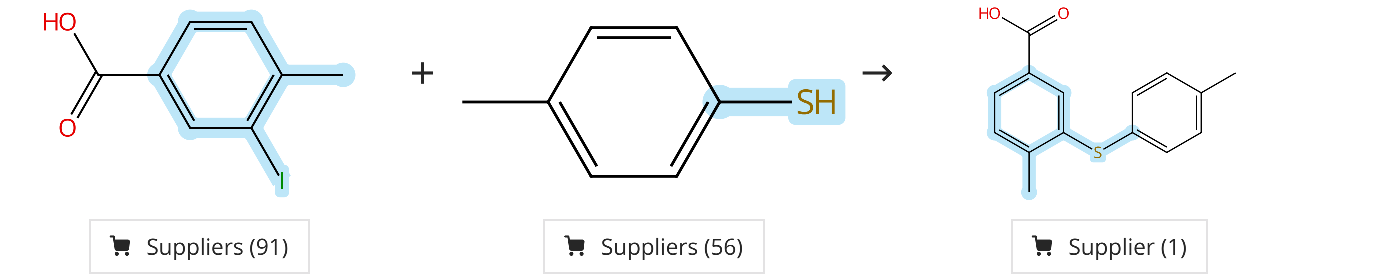
1.1 **Reagents:** Triethylamine, *N*-Methyl-2-pyrrolidone
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tris (dibenzylideneacetone)dipalladium; 21 h, 85 °C

Preparation of the Serotonin Transporter PET Radiotracer 2-({2-[(Dimethylamino)methyl]phenyl}thio)-5-[^{18}F]fluoroaniline (4-[^{18}F]ADAM): Probing Synthetic and Radiosynthetic Methods

By: Milicevic Sephton, Selena; et al

Synthesis (2019), 51(23), 4374-4384.

Scheme 53 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-10348490	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diisopropylcarbodiimide **Catalysts:** 4-(Dimethylamino)pyridine **Solvents:** Dimethylformamide, Dichloromethane; 15 h, 50 °C
1.2	**Reagents:** Amberlite IRA 400 (borohydride bound), Borohydride (bound to IRA 400) **Catalysts:** Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo- **Solvents:** Ethanol, Tetrahydrofuran; 15 h, 70 °C
1.3	**Reagents:** Trifluoroacetic acid **Solvents:** Dichloromethane; rt

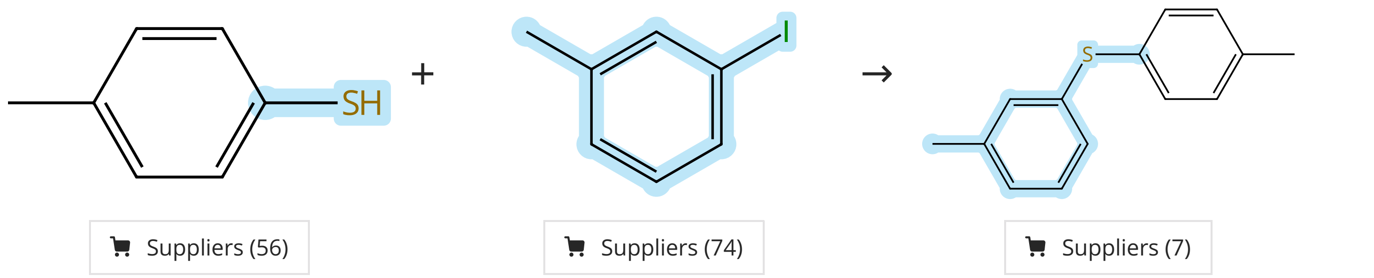
Experimental Protocols

Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 54 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-23873550	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Solvents:** Methanol, Polyethylene glycol; 15 min, 70 °C
1.2	**Catalysts:** Cuprous iodide; 12 h, 110 °C

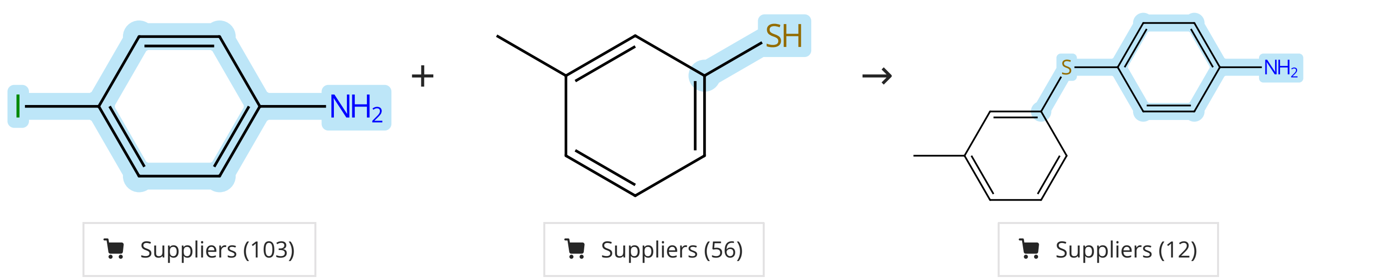
Experimental Protocols

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 55 (1 Reaction) Steps: **1** Yield: **100%**



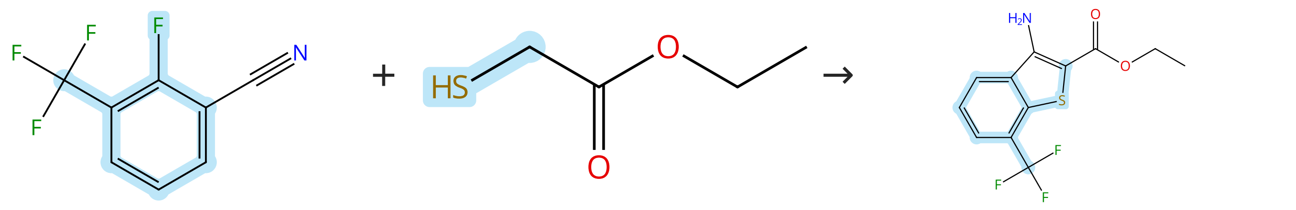
31-614-CAS-31632126	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cesium carbonate **Catalysts:** Copper oxide (CuO) **Solvents:** Toluene; 24 h, 110 °C
Experimental Protocols	

Rationale, synthesis and biological evaluation of substi tuted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 56 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (69) Suppliers (65) Suppliers (26)

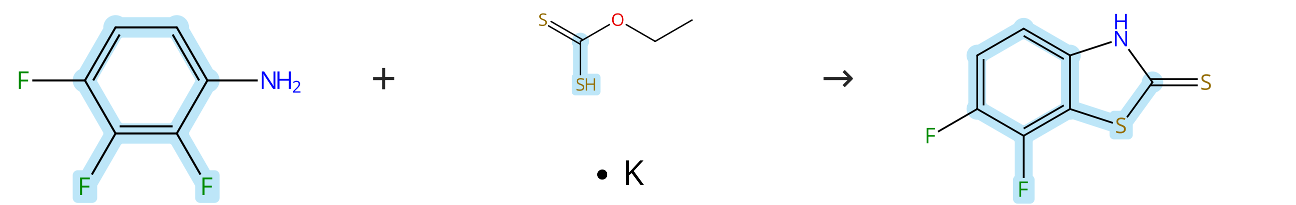
31-171-CAS-7912336	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Dimethylformamide; 0 °C; 0 °C → rt; 1.5 h, rt
1.2	**Reagents:** Water; 5 min, cooled

Synthesis and structure-activity relationship studies of 2-(1,3,4-oxadiazole-2(3H)-thione)-3-amino-5-arylthieno[2,3-b]pyridines as inhibitors of DRAK2

By: Leonczak, Piotr; et al

ChemMedChem (2014), 9(11), 2587-2601.

Scheme 57 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (72) Suppliers (70) Suppliers (17)

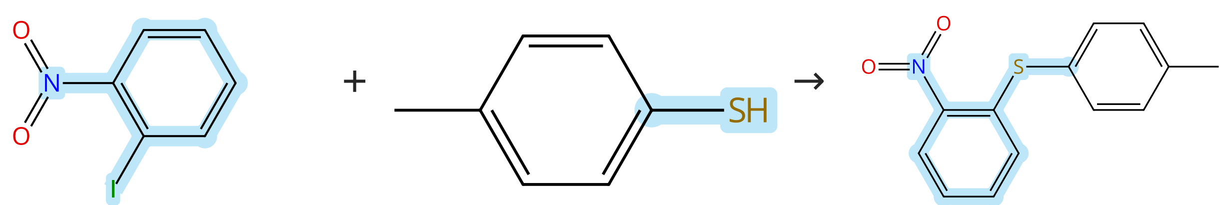
31-171-CAS-12578459	Steps: **1** Yield: **100%**
1.1	**Solvents:** Dimethylformamide; 1 h, 95 °C; 95 °C → rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 3 - 4, rt; 30 min, p H 3 - 4, rt
Experimental Protocols	

Ortho-Selective Nucleophilic Aromatic Substitution Reactions of Polyhaloanilines with Potassium/Sodium O-Ethyl Xanthate: A Convenient Access to Halogenated 2(3H)-Benzothiaz olethiones

By: Zhu, Lei; et al

Journal of Organic Chemistry (2004), 69(21), 7371-7374.

Scheme 58 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (67) Suppliers (56) Suppliers (8)

31-171-CAS-18817230	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Catalysts:** Bis(L-prolinato-κ*N*,κ*O*)palladium **Solvents:** Ethanol; 6 h, 70 °C

C-S Cross-Coupling Reaction Using a Recyclable Palladium Prolinate Catalyst under Mild and Green Conditions

By: Santos, Beatriz F.; et al

ChemistrySelect (2017), 2(28), 9063-9068.

Scheme 59 (1 Reaction) Steps: **1** Yield: **100%**

31-171-CAS-8309181	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cesium carbonate **Solvents:** Dimethyl sulfoxide; rt → 50 °C; 1 h, 50 °C; 50 °C → rt
1.2	**Reagents:** Sodium chloride **Solvents:** Water; rt

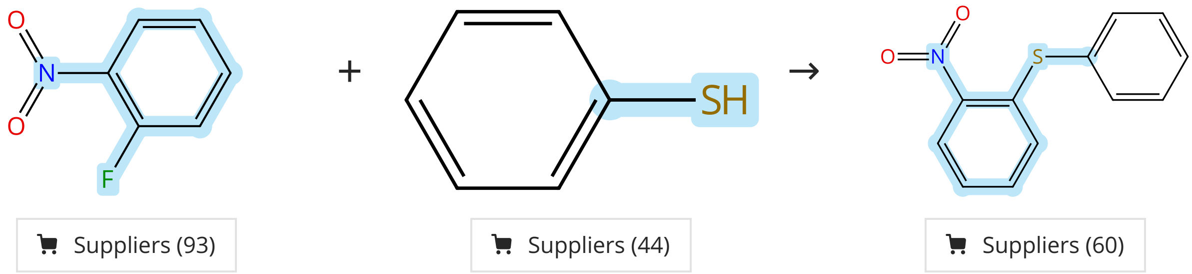
Experimental Protocols

An efficient protocol for synthesizing dibenzodithiapentalenes

By: Jepsen, Tue Heesgaard; et al

Tetrahedron Letters (2011), 52(31), 4045-4047.

Scheme 60 (1 Reaction) Steps: **1** Yield: **100%**



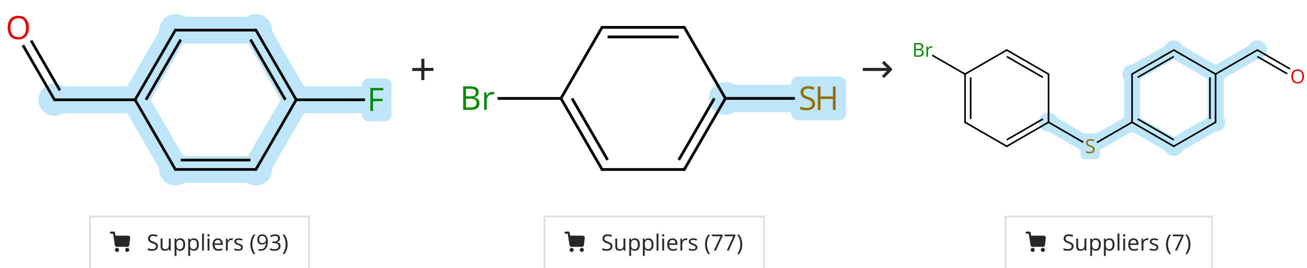
31-171-CAS-13679834	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 80 °C

Synthesis of substituted diphenyl sulfones and their structure-activity relationship with the antagonism of 5-HT6 receptors

By: Ivachtchenko, Alexandre; et al

Bioorganic & Medicinal Chemistry (2013), 21(15), 4614-4627.

Scheme 61 (1 Reaction) Steps: **1** Yield: **100%**



31-614-CAS-41211202	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 180 °C

Experimental Protocols

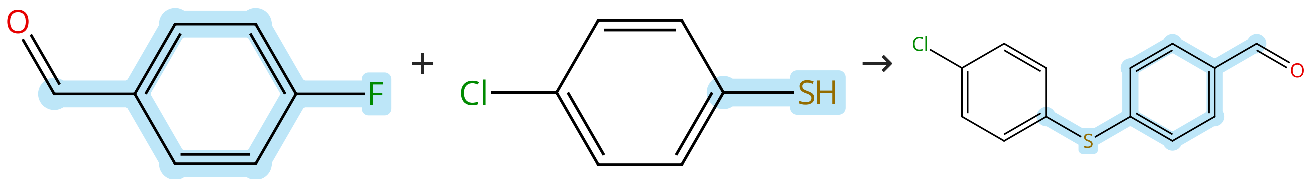
Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 62 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (93) | Suppliers (64) | Suppliers (21)

31-614-CAS-41211205

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

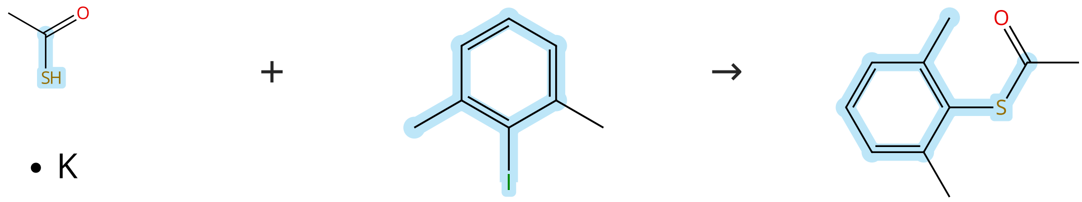
Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 63 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (84) | Suppliers (75) | Suppliers (4)

31-614-CAS-40340393

Steps: **1** Yield: **100%**

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C

Experimental Protocols

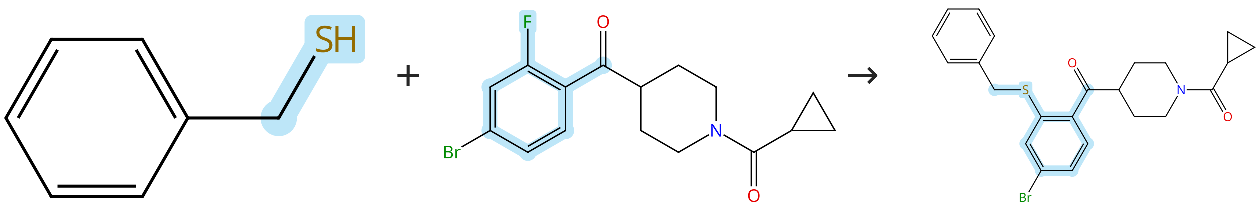
Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 64 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (53)

31-171-CAS-19716797

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; rt; 5 min, rt

1.2 **Solvents:** Tetrahydrofuran; 30 min, rt

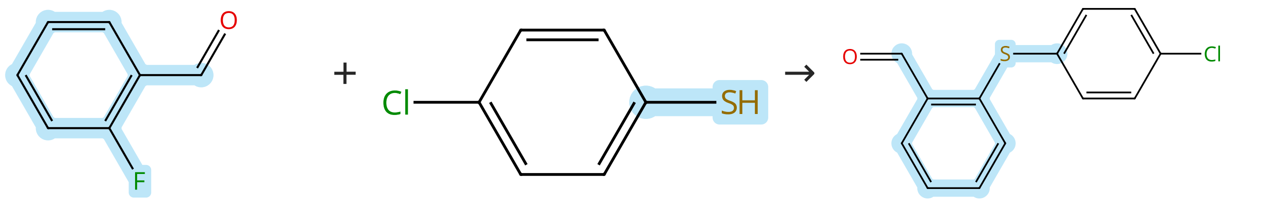
Experimental Protocols

Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 ($mGlu_4$)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 65 (3 Reactions) Steps: **1** Yield: **100%**



Suppliers (83) | Suppliers (64) | Suppliers (56)

31-614-CAS-38947796 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Intramolecular Friedel-Crafts Reaction with Trifluoroacetic Acid: Synthesizing Some New Functionalized 9-Aryl/Alkyl Thioxanthenes

By: Deniz, Hakan; et al

ACS Omega (2024), 9(11), 12596-12601.

31-614-CAS-36277489 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066778 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of new thioxanthenes by organocatalytic intramolecular Friedel-Crafts reaction

By: Yildiz, Tulay

Synthetic Communications (2018), 48(17), 2177-2188.

Scheme 66 (1 Reaction) Steps: **1** Yield: **100%**

31-614-CAS-35569400 Steps: **1** Yield: **100%**

1.1 **Solvents:** 1,4-Dioxane; 18 h, 100 °C

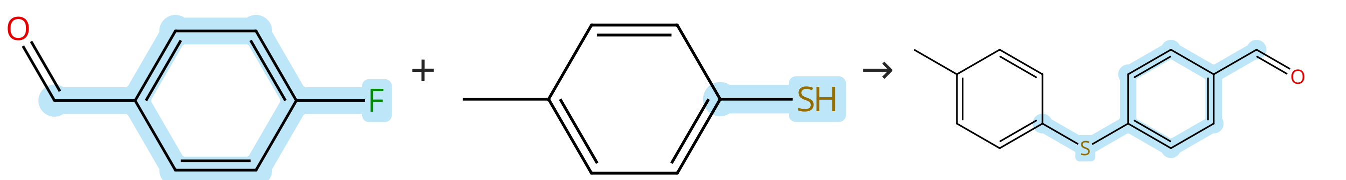
Experimental Protocols

Metal-Free S-Arylation of Phosphorothioate Diesters and Related Compounds with Diaryliodonium Salts

By: Sarkar, Sudeep; et al

Organic Letters (2023), 25(4), 671-675.

Scheme 67 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (93) | Suppliers (56) | Suppliers (31)

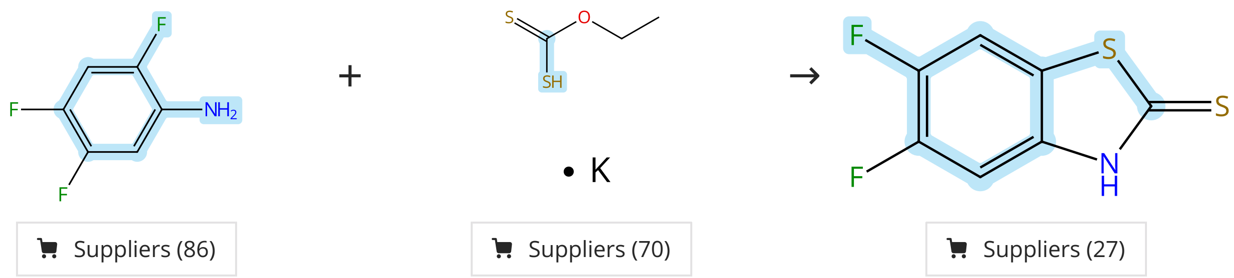
31-614-CAS-41211203	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 180 °C
Experimental Protocols	

Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 68 (1 Reaction) Steps: **1** Yield: **100%**



31-171-CAS-10478655	Steps: **1** Yield: **100%**
1.1	**Solvents:** Dimethylformamide; 2 h, 95 °C; 95 °C → rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 3 - 4, rt; 30 min, p H 3 - 4, rt
Experimental Protocols	

Ortho-Selective Nucleophilic Aromatic Substitution Reactions of Polyhaloanilines with Potassium/Sodium O-Ethyl Xanthate: A Convenient Access to Halogenated 2(3H)-Benzothiaz olethiones

By: Zhu, Lei; et al

Journal of Organic Chemistry (2004), 69(21), 7371-7374.

Scheme 69 (1 Reaction) Steps: **1** Yield: **100%**

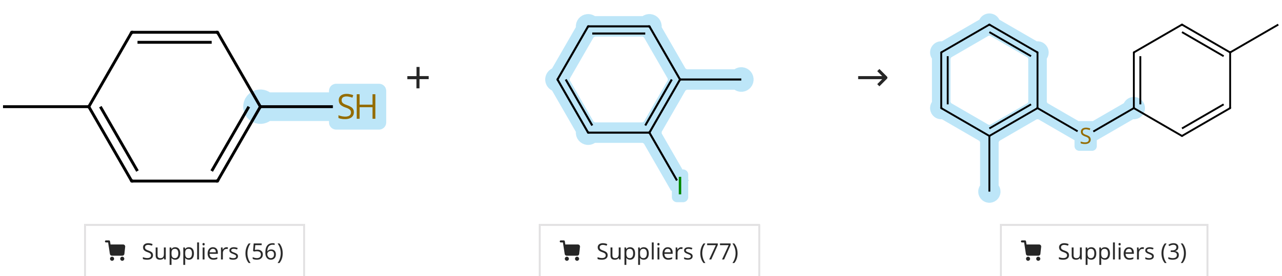
31-171-CAS-14141174	Steps: **1** Yield: **100%**
1.1	**Reagents:** 2,9-Dimethyl-1,10-phenanthroline, Sodium *tert*-butoxide **Solvents:** Toluene; 48 h, 110 °C
Experimental Protocols	

Copper- and palladium-containing perovskites: Catalysts for the Ullmann and Sonogashira reactions

By: Lohmann, Sophie; et al

Synlett (2005), (8), 1291-1295.

Scheme 70 (1 Reaction) Steps: **1** Yield: **100%**



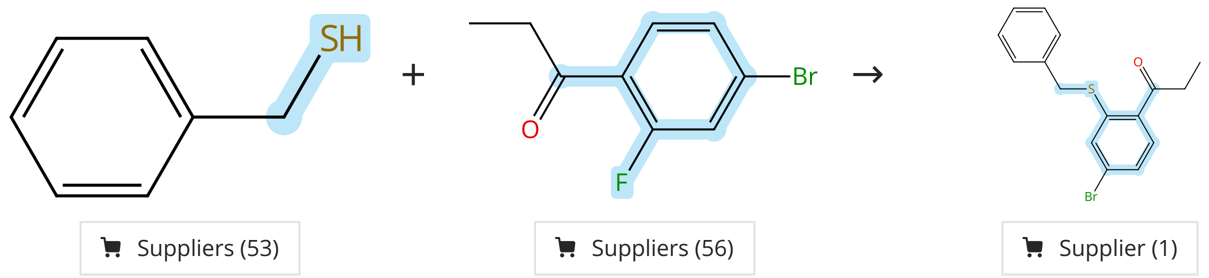
31-171-CAS-23871962	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Solvents:** Methanol, Polyethylene glycol; 15 min, 70 °C
1.2	**Catalysts:** Cuprous iodide; 12 h, 110 °C
Experimental Protocols	

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 71 (1 Reaction) Steps: **1** Yield: **100%**



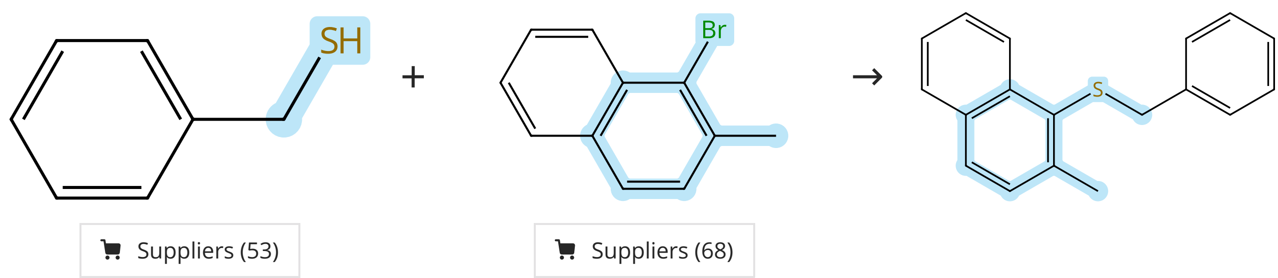
31-171-CAS-19716777	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; rt; 5 min, rt
1.2	**Solvents:** Tetrahydrofuran; 30 min, rt
Experimental Protocols	

Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 ($mGlu_4$)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 72 (1 Reaction) Steps: **1** Yield: **100%**



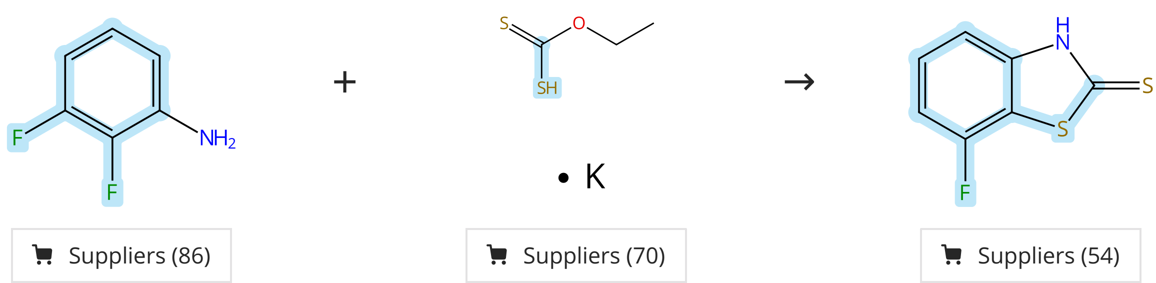
31-171-CAS-10748740	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium *tert*-butoxide **Solvents:** Tetrahydrofuran; 10 min, 23 °C
1.2	**Catalysts:** Zinc chloride **Solvents:** Tetrahydrofuran; 23 °C; 10 min, 23 °C
1.3	**Catalysts:** Lithium iodide, Di-μ-bromobis(tri-*tert*-butylphosphine)dipalladium **Solvents:** Tetrahydrofuran; 23 °C; 2 h, 90 °C
Experimental Protocols	

Zinc-Mediated Palladium-Catalyzed Formation of Carbon-Sulfur Bonds

By: Eichman, Chad C.; et al

Journal of Organic Chemistry (2009), 74(10), 4005-4008.

Scheme 73 (2 Reactions) Steps: **1** Yield: **100%**



31-171-CAS-10758451	Steps: **1** Yield: **100%**
1.1	**Solvents:** Dimethylformamide; 4 h, 100 °C; 100 °C → rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 5, rt
Experimental Protocols	

Small molecules enhance functional O-mannosylation of Alpha-dystroglycan

By: Lv, Fengping; et al

Bioorganic & Medicinal Chemistry (2015), 23(24), 7661-7670.

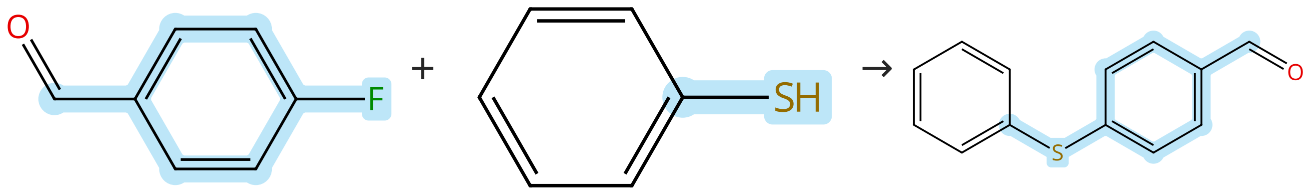
31-171-CAS-6208063	Steps: **1** Yield: **100%**
1.1	**Solvents:** Dimethylformamide; 2 h, 95 °C; 95 °C → rt
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 3 - 4, rt; 30 min, p H 3 - 4, rt
Experimental Protocols	

Ortho-Selective Nucleophilic Aromatic Substitution Reactions of Polyhaloanilines with Potassium/Sodium O-Ethyl Xanthate: A Convenient Access to Halogenated 2(3H)-Benzothiaz olethiones

By: Zhu, Lei; et al

Journal of Organic Chemistry (2004), 69(21), 7371-7374.

Scheme 74 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (93) Suppliers (44) Suppliers (55)

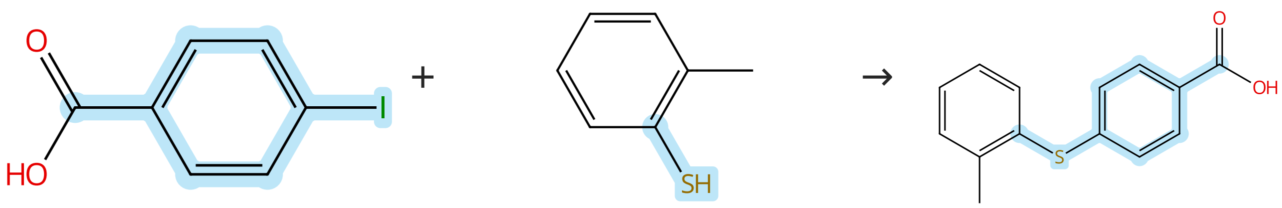
31-614-CAS-41211197	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 180 °C
Experimental Protocols	

Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 75 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (95) Suppliers (58) Suppliers (32)

31-171-CAS-7713082	Steps: **1** Yield: **100%**
1.1	**Reagents:** Diisopropylcarbodiimide **Catalysts:** 4-(Dimethylamino)pyridine **Solvents:** Dimethylformamide, Dichloromethane; 15 h, 50 °C
1.2	**Reagents:** Amberlite IRA 400 (borohydride bound), Borohy dride (bound to IRA 400) **Catalysts:** Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo- **Solvents:** Ethanol, Tetrahydrofuran; 15 h, 70 °C
1.3	**Reagents:** Trifluoroacetic acid **Solvents:** Dichloromethane; rt
Experimental Protocols	

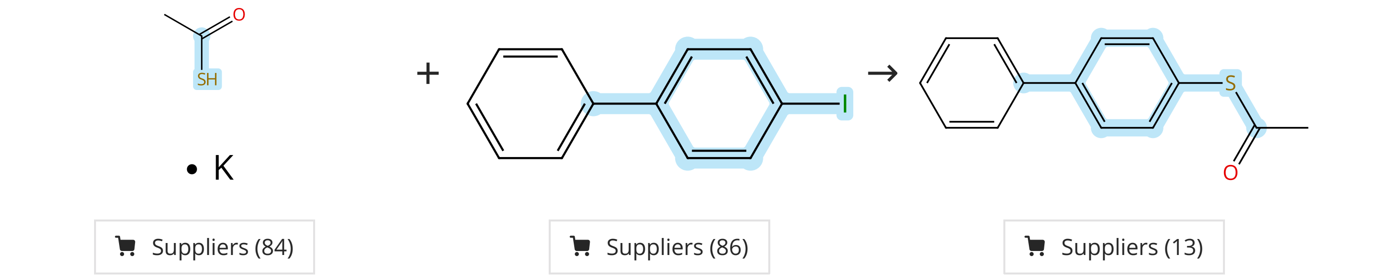
Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 76 (1 Reaction)

Steps: **1** Yield: **100%**



31-614-CAS-40340395

Steps: **1** Yield: **100%**

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C

Experimental Protocols

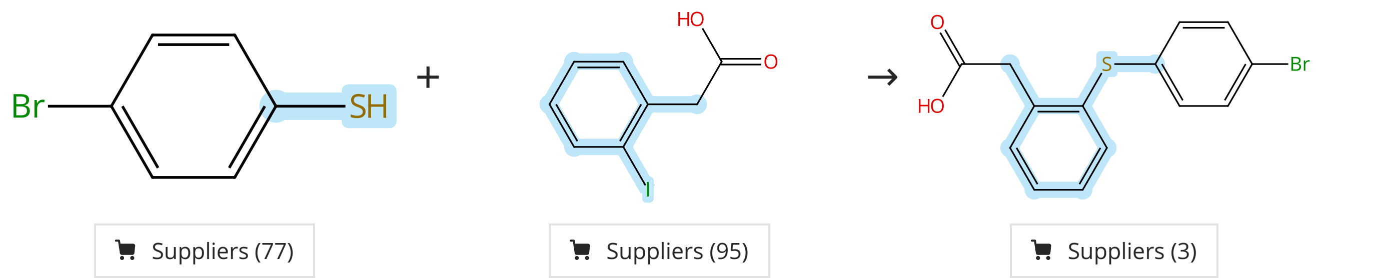
Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 77 (1 Reaction)

Steps: **1** Yield: **100%**



31-227-CAS-14254565

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Water; rt; rt → 50 °C

1.2 **Reagents:** Copper; 24 h, reflux

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; acidified, rt

Experimental Protocols

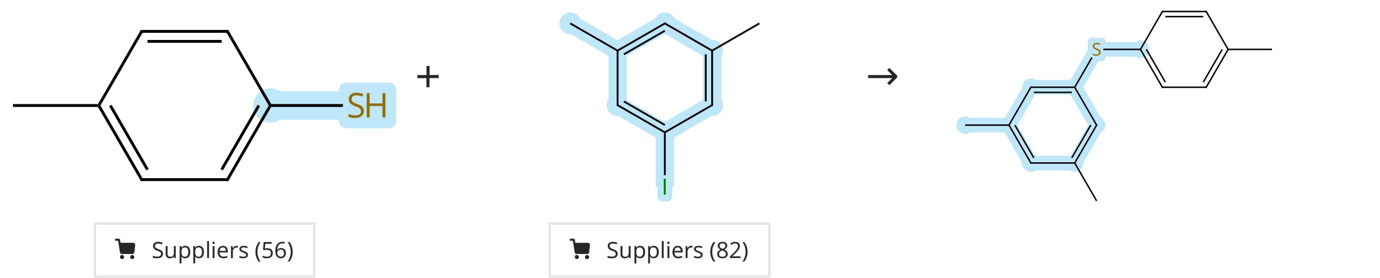
Exploring the Neuroleptic Substituent in Octoclothepin: Potential Ligands for Positron Emission Tomography with Subnanomolar Affinity for α_1-Adrenoceptors

By: Kristensen, Jesper L.; et al

Journal of Medicinal Chemistry (2010), 53(19), 7021-7034.

Scheme 78 (1 Reaction)

Steps: **1** Yield: **100%**



31-171-CAS-23873035

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C

1.2 **Catalysts:** Cuprous iodide; 12 h, 110 °C

Experimental Protocols

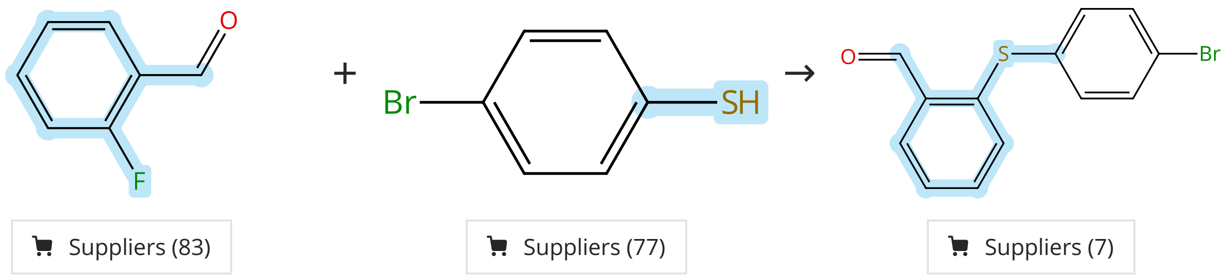
Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 79 (4 Reactions)

Steps: **1** Yield: **100%**



31-614-CAS-38947803

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Intramolecular Friedel-Crafts Reaction with Trifluoroacetic Acid: Synthesizing Some New Functionalized 9-Aryl/Alkyl Thioxanthenes

By: Deniz, Hakan; et al

ACS Omega (2024), 9(11), 12596-12601.

31-614-CAS-36277482

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066779

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of new thioxanthenes by organocatalytic intramo lecular Friedel-Crafts reaction

By: Yildiz, Tulay

Synthetic Communications (2018), 48(17), 2177-2188.

31-171-CAS-2812361

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

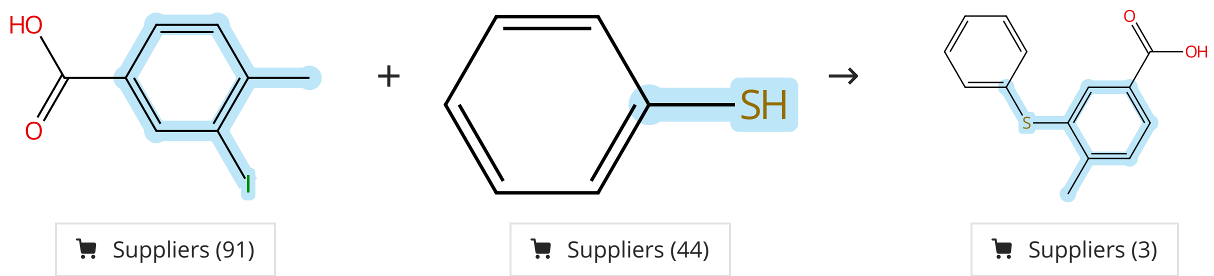
A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 80 (1 Reaction)

Steps: **1** Yield: **100%**



31-171-CAS-6304223

Steps: **1** Yield: **100%**

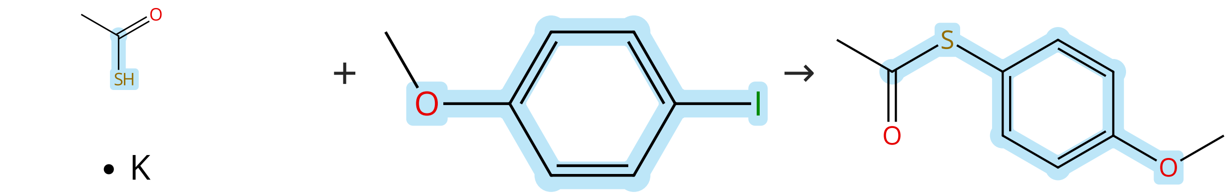
1.1 **Reagents:** Sodium hydroxide, Cuprous iodide
Solvents: Water

The synthesis of 3-hydroxymethyldibenzo[b,f]thiepin 5,5-dioxide, a prostaglandin antagonist

By: Hands, David; et al

Journal of Heterocyclic Chemistry (1986), 23(5), 1333-7.

Scheme 81 (1 Reaction) Steps: 1 Yield: 100%



Suppliers (84) Suppliers (89) Suppliers (18)

31-614-CAS-40340398 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; 2 h, 100 °C

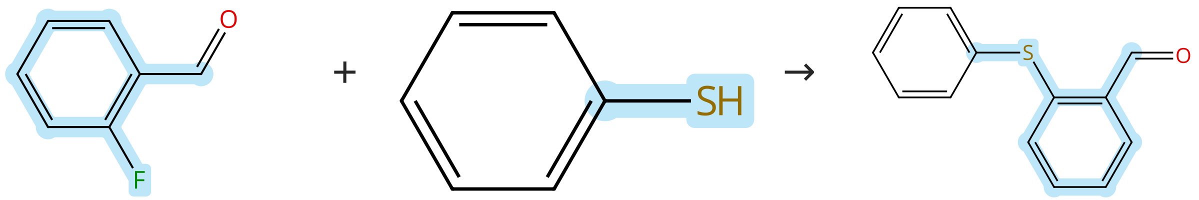
Experimental Protocols

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 82 (3 Reactions) Steps: 1 Yield: 100%



Suppliers (83) Suppliers (44) Suppliers (43)

31-614-CAS-36277487 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066775 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis of new thioxanthenes by organocatalytic intramo lecular Friedel-Crafts reaction

By: Yildiz, Tulay

Synthetic Communications (2018), 48(17), 2177-2188.

31-171-CAS-3114850 Steps: **1** Yield: **100%**

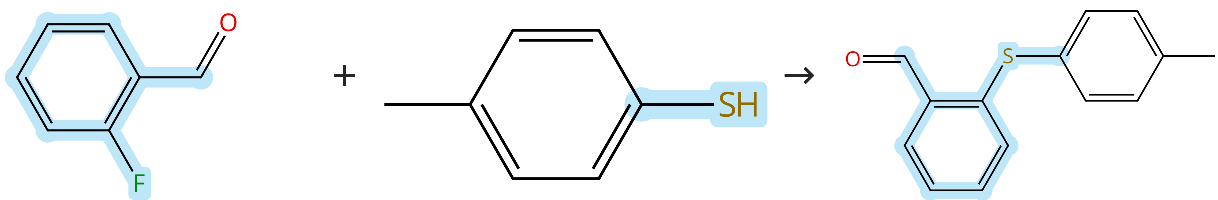
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 12 - 16 h, 50 - 80 °C

A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase

By: Zhang, Henry Q.; et al

Tetrahedron Letters (2003), 44(48), 8661-8663.

Scheme 83 (3 Reactions) Steps: 1 Yield: 100%



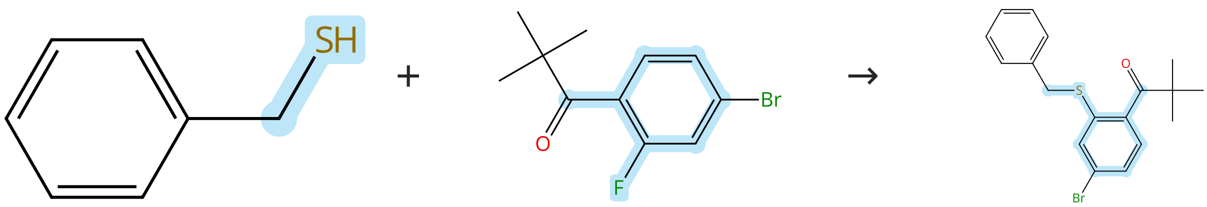
Suppliers (83) Suppliers (56) Suppliers (17)

31-614-CAS-36277484	Steps: **1** Yield: **100%**	Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 180 °C	By: Can Uskup, Hacer; et al
Experimental Protocols		ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066777	Steps: **1** Yield: **100%**	Synthesis of new thioxanthenes by organocatalytic intramo lecular Friedel-Crafts reaction
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 2 h, 180 °C	By: Yildiz, Tulay
Experimental Protocols		Synthetic Communications (2018), 48(17), 2177-2188.

31-171-CAS-98232	Steps: **1** Yield: **100%**	A concise synthesis of ortho-substituted aryl-acrylamides-potent activators of soluble guanylyl cyclase
1.1	**Reagents:** Potassium carbonate **Solvents:** Dimethylformamide; 12 - 16 h, 50 - 80 °C	By: Zhang, Henry Q.; et al
		Tetrahedron Letters (2003), 44(48), 8661-8663.

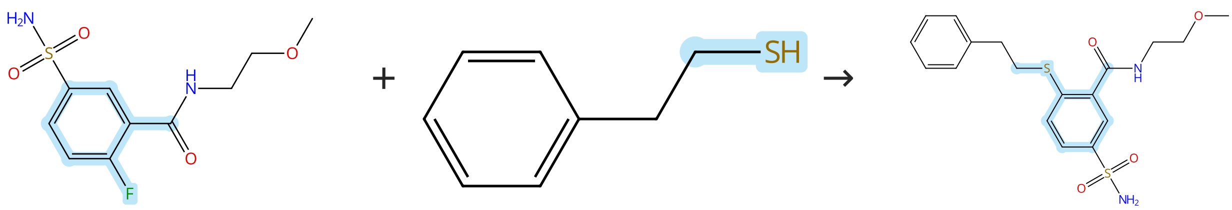
Scheme 84 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (53) Suppliers (35) Supplier (1)

31-171-CAS-19716779	Steps: **1** Yield: **100%**	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b] pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu$_4$)
1.1	**Reagents:** Potassium *tert*-butoxide **Solvents:** Tetrahydrofuran; rt; 5 min, rt	By: Bollinger, Sean R.; et al
1.2	**Solvents:** Tetrahydrofuran; 30 min, rt	Journal of Medicinal Chemistry (2019), 62(1), 342-358.
Experimental Protocols		

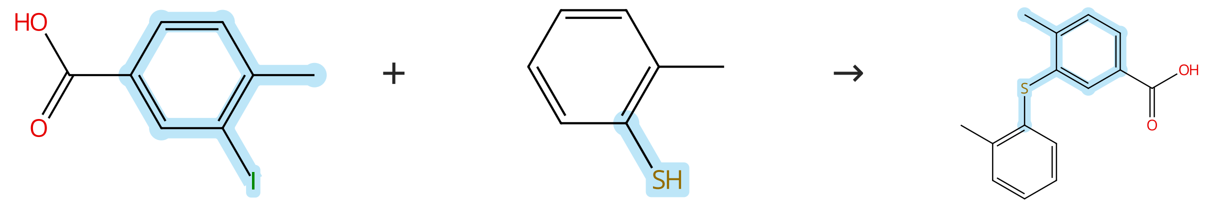
Scheme 85 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (4) Suppliers (53) Supplier (1)

31-171-CAS-1596308	Steps: **1** Yield: **100%**	Thioether benzenesulfonamide inhibitors of carbonic anhydrase II and IV: Structure-based drug design, synthesis, and biological evaluation
1.1	**Reagents:** Cesium carbonate **Solvents:** Dimethyl sulfoxide; 2 h, 50 - 100 °C	By: Vernier, William; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2010), 18(9), 3307-3319.

Scheme 86 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (91) | Suppliers (58) | Supplier (1)

31-171-CAS-12579220 Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dimethylformamide, Dichloromethane; 15 h, 50 °C

1.2 **Reagents:** Amberlite IRA 400 (borohydride bound), Borohydride (bound to IRA 400)
Catalysts: Nickel, bis(2,2'-bipyridine-κN^1,κ$N^{1'}$)dibromo-
Solvents: Ethanol, Tetrahydrofuran; 15 h, 70 °C

1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; rt

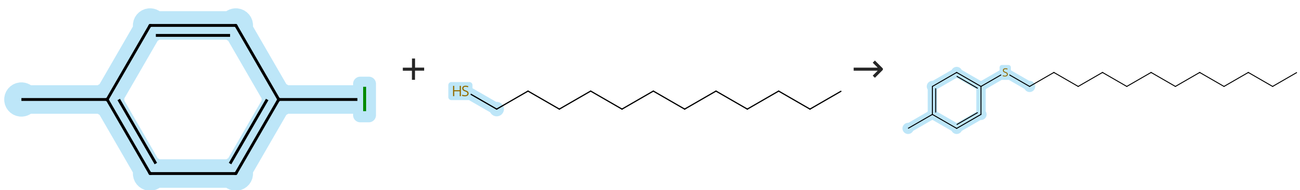
Experimental Protocols

Solid-Phase Synthesis of Diaryl Sulfides: Direct Coupling of Solid-Supported Aryl Halides with Thiols Using an Insoluble Polymer-Supported Reagent

By: Gendre, Fabrice; et al

Organic Letters (2005), 7(13), 2719-2722.

Scheme 87 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (82) | Suppliers (57) | Suppliers (3)

31-171-CAS-12266080 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium hydroxide
Catalysts: Copper oxide (Cu_2O)
Solvents: 1,4-Dioxane; 6 h, 110 °C

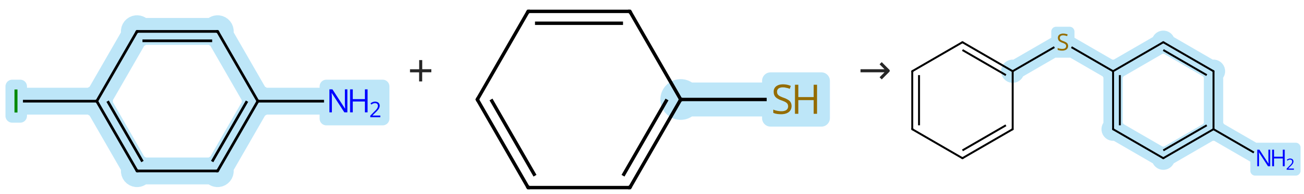
Experimental Protocols

An Efficient Copper-Catalyzed Cross-Coupling Reaction of Thiols with Aryl Iodides

By: Kao, Hsin-Lun; et al

European Journal of Organic Chemistry (2011), (9), 1776-1781, S1776/1-S1776/12.

Scheme 88 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (103) | Suppliers (44) | Suppliers (59)

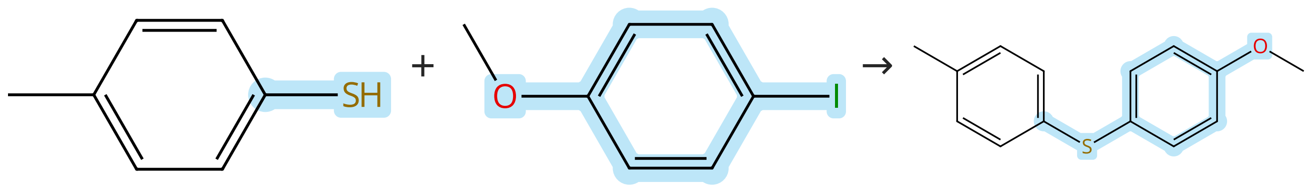
31-614-CAS-31632125	Steps: **1** Yield: **100%**
1.1	**Reagents:** Cesium carbonate **Catalysts:** Copper oxide (CuO) **Solvents:** Toluene; 24 h, 110 °C
Experimental Protocols	

Rationale, synthesis and biological evaluation of substi tuted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 89 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (56) Suppliers (89) Suppliers (4)

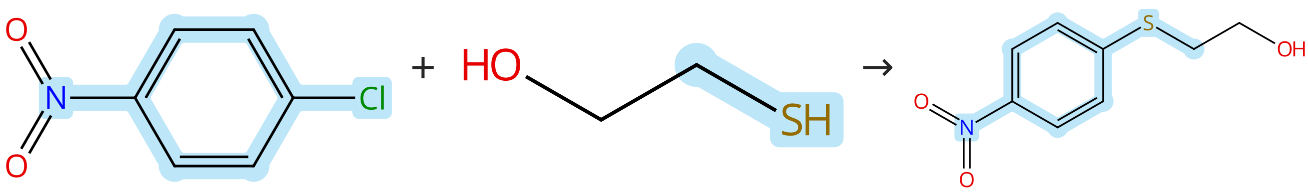
31-171-CAS-23875120	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide **Solvents:** Methanol, Polyethylene glycol; 15 min, 70 °C
1.2	**Catalysts:** Cuprous iodide; 12 h, 110 °C
Experimental Protocols	

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Scheme 90 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (65) Suppliers (118) Suppliers (48)

31-171-CAS-9391137	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium hydroxide

Nsc-mediated solid-phase synthesis of polyamides containing pyrrole amino acid

By: Choi, Jin Seok; et al

Tetrahedron Letters (2002), 43(24), 4295-4299.